

Memory and Long-Context Mechanisms for Long-Horizon LLM Agents: A Survey

Memory and long-context mechanisms are central to long-horizon LLM agents because they determine what an agent can remember, retrieve, compress, ignore, and justify across extended interaction trajectories. This survey reviews scholarly work on native long-context architectures, efficient attention, retrieval-augmented memory, episodic and semantic agent memory, context compression, cache management, long-horizon planning, and evaluation. We synthesize an arXiv-derived corpus of 300 papers collected during this run and organize the field around a systems view of memory: transient working context, persistent external stores, learned parametric knowledge, executable state, and auditable provenance. The survey highlights a recurring tension between capacity and control. Longer windows allow richer evidence but do not guarantee that models use relevant evidence; external memory improves persistence but introduces retrieval, freshness, privacy, and credit-assignment challenges. We close with prospects for memory hierarchies that combine scalable context, grounded retrieval, adaptive compression, and secure governance.

Keywords:

large language models; long-context modeling; memory; LLM agents; retrieval augmented generation; long-horizon planning

1 Introduction

Long-horizon LLM agents must operate over tasks that exceed a single prompt, a single document, or a single interaction. They may need to remember user preferences, maintain project state, revisit earlier observations, retrieve forgotten evidence, summarize long traces, and decide which past events matter for the next action. Memory and long-context mechanisms therefore form the substrate on which agent planning, tool use, personalization, and evaluation depend [10, 21, 66, 163, 184, 257, 262, 286]. Without a reliable memory layer, an agent can appear coherent locally while losing commitments, repeating work, or grounding decisions in stale context.

The field has developed along two complementary paths. One path increases the native context capacity of neural sequence models through efficient attention, positional extrapolation, cache management, recurrence, and long-sequence training [4, 10, 21, 60, 66, 163, 257, 293]. The other path builds external memory systems: retrieval indexes, vec-

tor databases, episodic logs, semantic stores, reflection buffers, and task-specific state representations [4, 10, 60, 66, 163, 184, 262, 293]. Long-horizon agents usually need both. A large context window can keep recent evidence visible, but persistent stores make experiences reusable across sessions; retrieval narrows the memory search space, but long-context reasoning is still needed to integrate the retrieved fragments.

This survey adopts a systems view. Memory is not a monolithic component; it is a hierarchy of representations with different latency, capacity, privacy, and reliability properties. Working context supports immediate reasoning, episodic memory stores interaction traces, semantic memory abstracts durable facts, procedural memory stores reusable skills, and external retrieval connects the agent to documents and databases. The research challenge is to coordinate these layers so that useful information remains accessible without overwhelming the model or violating user expectations.

2 Background

2.1 Memory Hierarchies for LLM Agents

Human-inspired terms such as working, episodic, and semantic memory are useful only when mapped to concrete system functions. In LLM agents, working memory is the active prompt or hidden state; episodic memory is a record of past episodes, observations, and actions; semantic memory contains distilled facts, user preferences, and domain knowledge; and procedural memory encodes reusable plans, tools, or skills. Long-horizon performance depends on transitions among these layers: what should be promoted from a trace to a durable summary, what should be retrieved for a new task, and what should be forgotten [4, 10, 60, 66, 163, 184, 262, 293].

2.2 The Long-Context Transformer Problem

Transformer-based LLMs face a basic scaling problem: attention over long sequences is expensive, and models trained on shorter contexts may not reliably use information placed far from the current token. Long-context research studies sparse or linear attention, recurrence, segment-level memory, positional encoding extrapolation, cache reuse, and training recipes that extend context windows [4, 10, 21, 60, 66, 163, 257, 293]. However, a longer window is not automatically a better memory. It changes the failure mode from missing information to misallocating attention, over-trusting irrelevant evidence, or failing to retrieve a detail buried in the middle of a prompt.

2.3 Retrieval as Externalized Context

Retrieval-augmented generation externalizes part of the context problem. Instead of placing all potentially relevant information in the prompt, the system indexes documents, memories, or tool outputs and retrieves a small working set at inference time [4, 10, 21, 60, 66, 163, 262, 293]. This strategy improves scalability and updateability, but it introduces new design questions: chunking, embedding choice, query rewriting, reranking, citation, conflict resolution, and freshness. For agents, retrieval is also temporal. The system must retrieve not only facts but prior goals, decisions, failed attempts, and constraints.

3 Methods

3.1 Search Protocol and Corpus Construction

The corpus for this survey was generated during this run from the arXiv structured API. Queries combined the task topic with memory, long context, context windows, long sequences, retrieval augmentation, episodic memory, semantic memory, agent memory, compression, KV cache, benchmarks, and long-horizon agent terms. Records were retained when title or abstract evidence connected language models or LLMs with memory, context extension, retrieval, agents, evaluation, training, or safety. Duplicates were removed by arXiv identifier, and the final bibliography cites 300 unique scholarly preprints.

3.2 Taxonomy

We organize methods along five axes. The first is *where* memory resides: model parameters, prompt context, retrieval indexes, databases, files, or environment state. The second is *when* memory is accessed: before planning, before each action, after observations, during reflection, or only at final answer time. The third is *how* memory is selected: dense retrieval, symbolic keys, recency, learned policies, salience scores, or human pinning. The fourth is *how* memory is transformed: raw traces, summaries, distilled facts, executable plans, or compressed hidden states. The fifth is *how* memory is governed: provenance, permissions, retention policies, privacy filters, and audit logs.

3.3 Native Long-Context Architectures

Native long-context methods attempt to increase the amount of evidence the model can directly condition on. Efficient attention and cache strategies reduce computational cost, while positional interpolation and extrapolation methods seek to preserve ordering information beyond the training length. These approaches are attractive because they avoid retrieval misses and allow cross-document synthesis inside the model’s active context. Their weakness is that capacity does not ensure use: models may still ignore distant evidence, overweight distractors, or fail to maintain temporal state over a long interaction [4, 10, 21, 60, 66, 163, 257, 293].

3.4 External Memory and Retrieval-Augmented Agents

External memory systems store information outside the model and expose it through retrieval or database calls. Agent memory differs from ordinary document retrieval because the stored objects are often temporal and action-centric: observations, plans, tool outputs, user corrections, environment states, and summaries of earlier attempts. Good memory systems therefore need schemas that preserve provenance, timestamps, task linkage, and permissions. Retrieval-augmented agents use these stores to maintain continuity across sessions, revisit evidence, and avoid recomputing prior work [4, 10, 21, 60, 66, 163, 262, 293].

3.5 Compression and Context Management

Long-horizon trajectories can grow faster than any context window. Compression methods summarize traces, merge redundant memories, distill facts, prune low-value turns, or maintain recurrent state. The design tension is lossy abstraction: compression improves tractability but may delete details needed for later reasoning. Context managers must therefore estimate future utility, not only current relevance. A promising direction is hierarchical memory, where raw logs remain available for audit while summaries and embeddings support fast control decisions [4, 36, 130, 137, 184, 209, 262, 286].

3.6 Memory for Planning and Long-Horizon Control

Planning agents need memory at multiple time scales. Short-term memory tracks the current subgoal and recent observation; medium-term memory tracks the task graph and unfinished commitments; long-term memory tracks user preferences, reusable skills, and domain-specific procedures. In long-horizon settings, failures often arise from memory-policy errors rather than reasoning errors: the agent forgets a constraint, retrieves the wrong episode, or treats a tentative plan as a fact. Research on agent memory therefore intersects with planning, reflection, tool use, and state estimation [10, 21, 66, 163, 184, 257, 262, 286].

3.7 Training and Adaptation

Memory policies can be hand-engineered, prompted, learned from demonstrations, or optimized through feedback. Training may teach models to write use-

ful memories, query external stores, compress traces, cite evidence, or decide when context is insufficient. Continual and personalized settings add another problem: the memory layer changes over time, so the agent must adapt without amplifying stale or incorrect information. Recent training work explores how long-context data, retrieval supervision, preference feedback, and reinforcement signals shape these behaviors [4, 77, 130, 137, 181, 249, 250, 293].

3.8 Benchmarks and Diagnostics

Evaluation for memory-rich agents must test more than answer accuracy. It should measure whether the system retrieves the right prior event, uses evidence at different positions in the context, handles distractors, maintains temporal order, updates memories after correction, and respects deletion or privacy requests. Long-context benchmarks and diagnostic tasks expose limitations such as position bias and lost-in-the-middle behavior, while agent benchmarks test whether memory supports multi-step task completion [4, 10, 21, 60, 66, 163, 286, 293].

3.9 Representative Literature Map

The following map cites the full retained corpus by theme, making the survey coverage explicit.

Long-context architectures and efficient attention

The long-context architectures and efficient attention cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [10, 60, 86, 95, 130, 184, 204, 257, 286]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [8, 18, 93, 112, 115, 135, 148, 190, 297]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [32, 67, 176, 177, 195, 228, 239, 251, 299]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [2, 56, 58, 71, 87, 107, 111, 157, 273]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [16, 126, 131, 136, 202, 227, 243, 245, 254]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [27, 33, 35, 43, 62, 121, 124, 129, 203]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [11, 42, 44, 46, 186, 265].

Retrieval-augmented memory and external stores

The retrieval-augmented memory and external stores cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [78, 85, 159, 169, 244, 262, 281, 289, 292]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [12, 25, 26, 45, 158, 166, 192, 193, 261]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [50, 68, 123, 168, 194, 238, 264, 277, 278]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [14, 23, 76, 81, 89, 144, 164, 196, 255]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [22, 79, 162, 183, 214, 218, 219, 230, 279]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [9, 28, 52, 96, 100, 103, 172, 225, 263]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [53, 63, 88, 109, 110, 222, 231, 242].

Episodic, semantic, and working memory for agents

The episodic, semantic, and working memory for agents cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [7, 36, 51, 77, 122, 145, 163, 181, 249]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [54, 147, 167, 170, 189, 232, 256, 259, 287]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [64, 134, 149, 156, 252, 269, 272, 288, 294]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [24, 65, 133, 140, 179, 206, 241, 258, 267]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [6, 90, 120, 175, 182, 207, 220, 240, 300].

Compression, summarization, and context management

The compression, summarization, and context management cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [20, 48, 113, 200, 208, 217, 236].

Planning and long-horizon agent control

The planning and long-horizon agent control cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [21, 37, 171, 173, 224, 226, 237, 250, 253]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [40, 49, 73, 74, 114, 142, 146, 285, 296].

Benchmarks, evaluation, and diagnostics

The benchmarks, evaluation, and diagnostics cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [4, 66, 82, 118, 137, 209, 212, 266, 293]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across datasets [19, 59, 72, 101, 160, 199, 210, 246, 295]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [55, 141, 143, 151, 185, 197, 213, 234, 283]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [5, 47, 99, 104, 127, 201, 233, 260, 271]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [38, 61, 98, 106, 125, 155, 205, 215, 276]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [30, 116, 187, 188, 223, 229, 274, 280, 282]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [1, 34, 41, 80, 94, 128, 161, 235, 248]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [70, 108, 117, 119, 174, 180, 198, 270, 290]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [15, 31, 57, 138, 139, 211, 247, 284, 298]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [3].

Training, adaptation, and reinforcement learning

The training, adaptation, and reinforcement learning cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [13, 39, 75, 83, 91, 97, 102, 165, 216]. A second strand in this cluster studies how such systems represent intermediate state, recover from failed calls, and compare tool-mediated behavior across

datasets [17, 29, 69, 92, 105, 132, 152, 153, 191]. Additional retained studies broaden the cluster with variants of interfaces, domains, training signals, and deployment assumptions [150, 178, 268, 275, 291].

Safety, provenance, and privacy of memory The safety, provenance, and privacy of memory cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [154, 221].

Multimodal and embodied long-horizon memory The multimodal and embodied long-horizon memory cluster contains work that treats LLMs as controllers of typed resources, stateful environments, or evaluation tasks [84].

4 Challenges and Prospects

4.1 Capacity Is Not Control

The most visible trend is larger context windows, but the core challenge is controlled use of context. Agents need to know which parts of memory are relevant, which are stale, which are uncertain, and which should be ignored. Future systems will combine long-context models with retrieval, salience estimation, and verification rather than treating window length as a complete solution.

4.2 Freshness, Conflict, and Provenance

Persistent memory creates conflicts: user preferences change, retrieved documents disagree, and prior summaries may be wrong. Agents need provenance meta-data that links each memory to its source, time, confidence, and permissions. They also need mechanisms for correction and deletion. Without these controls, memory can turn isolated hallucinations into durable state [51, 54, 135, 160, 210, 224, 237, 257].

4.3 Privacy and User Agency

Long-term memory increases personalization but also increases privacy risk. Users need to understand what is stored, why it is retrieved, how long it persists, and how to remove it. Technical work on memory should therefore be paired with interface and governance work: inspectable memory stores, consent-aware retention, task-scoped permissions, and audit logs.

4.4 Evaluation Under Realistic Horizons

Many memory benchmarks compress the problem into synthetic retrieval tasks. Such diagnostics are useful, but long-horizon agents also need evaluations with changing goals, noisy observations, tool failures, and delayed consequences. The field needs versioned environments and trace-level metrics that distinguish retrieval failure, compression failure, reasoning failure, and unsafe memory use.

5 Conclusion

Memory and long-context mechanisms determine whether LLM agents can act coherently over extended tasks. This survey reviewed native long-context architectures, retrieval-augmented memory, episodic and semantic stores, compression, planning memory, training signals, benchmarks, and governance. The central lesson is that memory is a controlled systems interface, not merely extra tokens. Future long-horizon agents will need memory hierarchies that are scalable, grounded, auditable, and aligned with user intent.

References

- [1] Lukas Aichberger and Sepp Hochreiter. Unlocking the Working Memory of Large Language Models for Latent Reasoning. *arXiv preprint arXiv:2605.30343*, 2026. URL <https://arxiv.org/abs/2605.30343>.
- [2] Chenxin An, Fei Huang, Jun Zhang, Shansan Gong, Xipeng Qiu, Chang Zhou, and Lingpeng Kong. Training-Free Long-Context Scaling of Large Language Models. *arXiv preprint arXiv:2402.17463*, 2024. URL <https://arxiv.org/abs/2402.17463>.
- [3] Mayank Arya and Yogesh Simmhan. Understanding the Performance and Power of LLM Inference on Edge Accelerators. *arXiv preprint arXiv:2506.09554*, 2025. URL <https://arxiv.org/abs/2506.09554>.
- [4] Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. LongBench: A Bilingual, Multitask Benchmark for Long Context Understanding. *arXiv preprint arXiv:2308.14508*, 2023. URL <https://arxiv.org/abs/2308.14508>.
- [5] Yushi Bai, Xin Lv, Jiajie Zhang, Yuze He, Ji Qi, Lei Hou, Jie Tang, Yuxiao Dong, and Juanzi Li.

- LongAlign: A Recipe for Long Context Alignment of Large Language Models. *arXiv preprint arXiv:2401.18058*, 2024. URL <https://arxiv.org/abs/2401.18058>.
- [6] Ondrej Bohdal, Pramit Saha, Umberto Michieli, Mete Ozay, and Taha Ceritli. Clustering-driven Memory Compression for On-device Large Language Models. *arXiv preprint arXiv:2601.17443*, 2026. URL <https://arxiv.org/abs/2601.17443>.
- [7] Luiz C. Borro, Luiz A. B. Macarini, Gordon Tindall, Michael Montero, and Adam B. Struck. Memori: A Persistent Memory Layer for Efficient, Context-Aware LLM Agents. *arXiv preprint arXiv:2603.19935*, 2026. URL <https://arxiv.org/abs/2603.19935>.
- [8] Ruisi Cai, Yuandong Tian, Zhangyang Wang, and Beidi Chen. LoCoCo: Dropping In Convolutions for Long Context Compression. *arXiv preprint arXiv:2406.05317*, 2024. URL <https://arxiv.org/abs/2406.05317>.
- [9] Jiaqi Cao, Jiarui Wang, Rubin Wei, Qipeng Guo, Kai Chen, Bowen Zhou, and Zhouhan Lin. Memory Decoder: A Pretrained, Plug-and-Play Memory for Large Language Models. *arXiv preprint arXiv:2508.09874*, 2025. URL <https://arxiv.org/abs/2508.09874>.
- [10] Weili Cao, Xunjian Yin, Bhuwan Dhingra, and Shuyan Zhou. Coding Agents are Effective Long-Context Processors. *arXiv preprint arXiv:2603.20432*, 2026. URL <https://arxiv.org/abs/2603.20432>.
- [11] Arie Cattan, Alon Jacovi, Alex Fabrikant, Jonathan Herzig, Roei Aharoni, Hannah Rashkin, Dror Marcus, Avinatan Hassidim, Yossi Matias, Idan Szpektor, and Avi Caciularu. DoubleDipper: Improving Long-Context LLMs via Context Recycling. *arXiv preprint arXiv:2406.13632*, 2024. URL <https://arxiv.org/abs/2406.13632>.
- [12] Guanzheng Chen, Qilong Feng, Jinjie Ni, Xin Li, and Michael Qizhe Shieh. RAPID: Long-Context Inference with Retrieval-Augmented Speculative Decoding. *arXiv preprint arXiv:2502.20330*, 2025. URL <https://arxiv.org/abs/2502.20330>.
- [13] Guanzheng Chen, Xin Li, Michael Qizhe Shieh, and Lidong Bing. LongPO: Long Context Self-Evolution of Large Language Models through Short-to-Long Preference Optimization. *arXiv preprint arXiv:2502.13922*, 2025. URL <https://arxiv.org/abs/2502.13922>.
- [14] Jianghao Chen, Junhong Wu, Yangyifan Xu, and Jiajun Zhang. LADM: Long-context Training Data Selection with Attention-based Dependency Measurement for LLMs. *arXiv preprint arXiv:2503.02502*, 2025. URL <https://arxiv.org/abs/2503.02502>.
- [15] Jinbiao Chen, Shuang Jin, Guoyun Zhang, Junyu Zhang, Guanyi Wang, and Hanzhang Qin. Automated Reformulation of Robust Optimization via Memory-Augmented Large Language Models. *arXiv preprint arXiv:2605.11813*, 2026. URL <https://arxiv.org/abs/2605.11813>.
- [16] Longze Chen, Ziqiang Liu, Wanwei He, Yunshui Li, Run Luo, and Min Yang. Long Context is Not Long at All: A Prospector of Long-Dependency Data for Large Language Models. *arXiv preprint arXiv:2405.17915*, 2024. URL <https://arxiv.org/abs/2405.17915>.
- [17] Yiming Chen, Yuan Zhang, Yin Liu, Kun Yuan, and Zaiwen Wen. A Memory Efficient Randomized Subspace Optimization Method for Training Large Language Models. *arXiv preprint arXiv:2502.07222*, 2025. URL <https://arxiv.org/abs/2502.07222>.
- [18] Yukang Chen, Shengju Qian, Haotian Tang, Xin Lai, Zhijian Liu, Song Han, and Jiaya Jia. LongLoRA: Efficient Fine-tuning of Long-Context Large Language Models. *arXiv preprint arXiv:2309.12307*, 2023. URL <https://arxiv.org/abs/2309.12307>.
- [19] Ziyang Chen, Xing Wu, Junlong Jia, Chaochen Gao, Qi Fu, Debing Zhang, and Songlin Hu. LongBench Pro: A More Realistic and Comprehensive Bilingual Long-Context Evaluation Benchmark. *arXiv preprint arXiv:2601.02872*, 2026. URL <https://arxiv.org/abs/2601.02872>.
- [20] Hongrong Cheng, Miao Zhang, and Javen Qin-feng Shi. MINI-LLM: Memory-Efficient Structured Pruning for Large Language Models. *arXiv preprint arXiv:2407.11681*, 2024. URL <https://arxiv.org/abs/2407.11681>.
- [21] Andy Chung, Yichi Zhang, Kaixiang Lin, Aditya Rawal, Qiaozi Gao, and Joyce Chai. Evaluating Long-Context Reasoning in LLM-Based WebAgents. *arXiv preprint arXiv:2512.04307*, 2025. URL <https://arxiv.org/abs/2512.04307>.
- [22] Michael Cuccarese. Epistemic Blinding: An Inference-Time Protocol for Auditing Prior Contamination in LLM-Assisted Analysis. *arXiv preprint arXiv:2604.06013*, 2026. URL <https://arxiv.org/abs/2604.06013>.
- [23] Payel Das, Subhajit Chaudhury, Elliot Nelson, Igor Melnyk, Sarath Swaminathan, Sihui Dai, Aurelie Lozano, Georgios Kollias, Vijil Chenthamarakshan, Jiri, Navratil, Soham Dan, and Pin-Yu Chen. Larimar: Large Language Models with Episodic Memory Control. *arXiv preprint arXiv:2403.11901*, 2024. URL <https://arxiv.org/abs/2403.11901>.
- [24] Pritam Dash, Tongyu Ge, Aditi Jain, Tanmay Shah, and Zhiwei Shang. From Untrusted Input to Trusted Memory: A Systematic Study of Memory

- Poisoning Attacks in LLM Agents. *arXiv preprint arXiv:2606.04329*, 2026. URL <https://arxiv.org/abs/2606.04329>.
- [25] Jingyang Deng, Zhengyang Shen, Boyang Wang, Lixin Su, Suqi Cheng, Ying Nie, Junfeng Wang, Dawei Yin, and Jinwen Ma. FltLM: An Intergrated Long-Context Large Language Model for Effective Context Filtering and Understanding. *arXiv preprint arXiv:2410.06886*, 2024. URL <https://arxiv.org/abs/2410.06886>.
- [26] Xinle Deng, Ruobin Zhong, Hujin Peng, Xiaoben Lu, Yanzhe Wu, Guang Li, Buqiang Xu, Yunzhi Yao, Jizhan Fang, Haoliang Cao, Junjie Guo, Yuan Yuan, Ziqing Ma, Yuanqiang Yu, Rui Hu, Baohua Dong, Hangcheng Zhu, and Ningyu Zhang. MemTrace: Tracing and Attributing Errors in Large Language Model Memory Systems. *arXiv preprint arXiv:2605.28732*, 2026. URL <https://arxiv.org/abs/2605.28732>.
- [27] Yangshen Deng, Zhengxin You, Long Xiang, Qilong Li, Peiqi Yuan, Zhaoyang Hong, Yitao Zheng, Wanting Li, Runzhong Li, Haotian Liu, Kyriakos Mouratidis, Man Lung Yiu, Huan Li, Qiaomu Shen, Rui Mao, and Bo Tang. AlayaDB: The Data Foundation for Efficient and Effective Long-context LLM Inference. *arXiv preprint arXiv:2504.10326*, 2025. URL <https://arxiv.org/abs/2504.10326>.
- [28] Youwang Deng. Substrate Asymmetry in User-Side Memory: A Diagnostic Framework. *arXiv preprint arXiv:2606.11712*, 2026. URL <https://arxiv.org/abs/2606.11712>.
- [29] Dai Do, Quan Tran, Svetha Venkatesh, and Hung Le. Large Language Models Prompting With Episodic Memory. *arXiv preprint arXiv:2408.07465*, 2024. URL <https://arxiv.org/abs/2408.07465>.
- [30] Quyet V. Do, Thinh Pham, Nguyen Nguyen, Sha Li, Pratibha Zunjare, and Tu Vu. $\2 : Structure-Originated Reasoning Data Improves Long-Context Reasoning Ability of Large Language Models. *arXiv preprint arXiv:2604.05114*, 2026. URL <https://arxiv.org/abs/2604.05114>.
- [31] Songwei Dong, Zihan Chen, Chengshuai Shi, Peng Wang, Jundong Li, and Cong Shen. Is One Score Enough? Rethinking the Evaluation of Sequentially Evolving LLM Memory. *arXiv preprint arXiv:2605.15384*, 2026. URL <https://arxiv.org/abs/2605.15384>.
- [32] Zican Dong, Junyi Li, Jinhao Jiang, Mingyu Xu, Wayne Xin Zhao, Bingning Wang, and Weipeng Chen. LongReD: Mitigating Short-Text Degradation of Long-Context Large Language Models via Restoration Distillation. *arXiv preprint arXiv:2502.07365*, 2025. URL <https://arxiv.org/abs/2502.07365>.
- [33] Junjie Fang, Likai Tang, Hongzhe Bi, Yujia Qin, Si Sun, Zhenyu Li, Haolun Li, Yongjian Li, Xin Cong, Yankai Lin, Yukun Yan, Xiaodong Shi, Sen Song, Zhiyuan Liu, and Maosong Sun. UniMem: Towards a Unified View of Long-Context Large Language Models. *arXiv preprint arXiv:2402.03009*, 2024. URL <https://arxiv.org/abs/2402.03009>.
- [34] Lizhe Fang, Yifei Wang, Zhaoyang Liu, Chenheng Zhang, Stefanie Jegelka, Jinyang Gao, Bolin Ding, and Yisen Wang. What is Wrong with Perplexity for Long-context Language Modeling? *arXiv preprint arXiv:2410.23771*, 2024. URL <https://arxiv.org/abs/2410.23771>.
- [35] Yunhua Fang, Rui Xie, Asad Ul Haq, Linsen Ma, Kaoutar El Maghraoui, Naigang Wang, Meng Wang, Liu Liu, and Tong Zhang. Accelerating LLM Inference via Dynamic KV Cache Placement in Heterogeneous Memory System. *arXiv preprint arXiv:2508.13231*, 2025. URL <https://arxiv.org/abs/2508.13231>.
- [36] Tao Feng, Chongrui Ye, Tianyang Luo, Jingjun Xu, Xueqiang Xu, Haozhen Zhang, Ge Liu, and Jiaxuan You. ElasticMem: Latent Memory as a Learnable Resource for LLM Agents. *arXiv preprint arXiv:2605.30690*, 2026. URL <https://arxiv.org/abs/2605.30690>.
- [37] Tao Feng, Haozhen Zhang, Zijie Lei, Peixuan Han, and Jiaxuan You. GraphPlanner: Graph Memory-Augmented Agentic Routing for Multi-Agent LLMs. *arXiv preprint arXiv:2604.23626*, 2026. URL <https://arxiv.org/abs/2604.23626>.
- [38] Jeffrey Flynt. Structured Belief State and the First Precision-Aware Benchmark for LLM Memory Retrieval. *arXiv preprint arXiv:2605.11325*, 2026. URL <https://arxiv.org/abs/2605.11325>.
- [39] Saman Forouzandeh, Wei Peng, Parham Moradi, Xinghuo Yu, and Mahdi Jalili. Learning Hierarchical Procedural Memory for LLM Agents through Bayesian Selection and Contrastive Refinement. *arXiv preprint arXiv:2512.18950*, 2025. URL <https://arxiv.org/abs/2512.18950>.
- [40] Kazuki Fujii, Kohei Watanabe, and Rio Yokota. Accelerating Large Language Model Training with 4D Parallelism and Memory Consumption Estimator. *arXiv preprint arXiv:2411.06465*, 2024. URL <https://arxiv.org/abs/2411.06465>.
- [41] Victor Gallego. Distilling Feedback into Memory-as-a-Tool. *arXiv preprint arXiv:2601.05960*, 2026. URL <https://arxiv.org/abs/2601.05960>.
- [42] Chaochen Gao, Xing Wu, Qi Fu, and Songlin Hu. Quest: Query-centric Data Synthesis Approach for Long-context Scaling of Large Language Model. *arXiv preprint arXiv:2405.19846*, 2024. URL <https://arxiv.org/abs/2405.19846>.

- [43] Chaochen Gao, Xing Wu, Zijia Lin, Debing Zhang, and Songlin Hu. LongMagpie: A Self-synthesis Method for Generating Large-scale Long-context Instructions. *arXiv preprint arXiv:2505.17134*, 2025. URL <https://arxiv.org/abs/2505.17134>.
- [44] Chaochen Gao, Xing Wu, Zijia Lin, Debing Zhang, and Songlin Hu. NExtLong: Toward Effective Long-Context Training without Long Documents. *arXiv preprint arXiv:2501.12766*, 2025. URL <https://arxiv.org/abs/2501.12766>.
- [45] Hang Gao and Yongfeng Zhang. INMS: Memory Sharing for Large Language Model based Agents. *arXiv preprint arXiv:2404.09982*, 2024. URL <https://arxiv.org/abs/2404.09982>.
- [46] Yifei Gao, Shaohong Chen, Lei Wang, Ruiting Dai, Ziyun Zhang, Kerui Ren, Jiaji Wu, and Jun Cheng. Pay Attention to What You Need. *arXiv preprint arXiv:2307.13365*, 2023. URL <https://arxiv.org/abs/2307.13365>.
- [47] Ishir Garg, Neel Kolhe, Dawn Song, and Xuan-dong Zhao. MemFail: Stress-Testing Failure Modes of LLM Memory Systems. *arXiv preprint arXiv:2605.26667*, 2026. URL <https://arxiv.org/abs/2605.26667>.
- [48] Tao Ge, Jing Hu, Lei Wang, Xun Wang, Si-Qing Chen, and Furu Wei. In-context Autoencoder for Context Compression in a Large Language Model. *arXiv preprint arXiv:2307.06945*, 2023. URL <https://arxiv.org/abs/2307.06945>.
- [49] Aditi Godbole, Jabin Geevarghese George, and Smita Shandilya. Leveraging Long-Context Large Language Models for Multi-Document Understanding and Summarization in Enterprise Applications. *arXiv preprint arXiv:2409.18454*, 2024. URL <https://arxiv.org/abs/2409.18454>.
- [50] Jiani Guo, Zuchao Li, Jie Wu, Qianren Wang, Yun Li, Lefei Zhang, Hai Zhao, and Yujiu Yang. ToM: Leveraging Tree-oriented MapReduce for Long-Context Reasoning in Large Language Models. *arXiv preprint arXiv:2511.00489*, 2025. URL <https://arxiv.org/abs/2511.00489>.
- [51] Jing Guo, Nan Li, Jianchuan Qi, Hang Yang, Ruiqiao Li, Yuzhen Feng, Si Zhang, and Ming Xu. Empowering Working Memory for Large Language Model Agents. *arXiv preprint arXiv:2312.17259*, 2023. URL <https://arxiv.org/abs/2312.17259>.
- [52] Bernal Jimenez Gutierrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. HippoRAG: Neurobiologically Inspired Long-Term Memory for Large Language Models. *arXiv preprint arXiv:2405.14831*, 2024. URL <https://arxiv.org/abs/2405.14831>.
- [53] Bernal Jimenez Gutierrez, Yiheng Shu, Weijian Qi, Sizhe Zhou, and Yu Su. From RAG to Memory: Non-Parametric Continual Learning for Large Language Models. *arXiv preprint arXiv:2502.14802*, 2025. URL <https://arxiv.org/abs/2502.14802>.
- [54] Hyeonjeong Ha, Jeonghwan Kim, Cheng Qian, Jiayu Liu, William M. Campbell, Yue Wu, Yuji Zhang, Kathleen McKeown, Dilek Hakkani-Tur, and Heng Ji. MemGuard: Preventing Memory Contamination in Long-Term Memory-Augmented Large Language Models. *arXiv preprint arXiv:2605.28009*, 2026. URL <https://arxiv.org/abs/2605.28009>.
- [55] Xinjie He, Zhiyuan Lin, Su Liu, Jialun Wu, Qiyang Xie, Weikai Zhou, and Shuai Xiao. What Training Data Teaches RL Memory Agents: An Empirical Study of Curriculum Effects in Memory-Augmented QA. *arXiv preprint arXiv:2605.23067*, 2026. URL <https://arxiv.org/abs/2605.23067>.
- [56] Zexue He, Leonid Karlinsky, Donghyun Kim, Julian McAuley, Dmitry Krotov, and Rogerio Feris. CAMELoT: Towards Large Language Models with Training-Free Consolidated Associative Memory. *arXiv preprint arXiv:2402.13449*, 2024. URL <https://arxiv.org/abs/2402.13449>.
- [57] Zhuomin He, Yizhen Yao, Pengfei Zuo, Bin Gao, Qinya Li, Zhenzhe Zheng, and Fan Wu. AdaSkip: Adaptive Sublayer Skipping for Accelerating Long-Context LLM Inference. *arXiv preprint arXiv:2501.02336*, 2025. URL <https://arxiv.org/abs/2501.02336>.
- [58] Zifan He, Rui Ma, Yizhou Sun, and Jason Cong. Understand and Accelerate Memory Processing Pipeline for Large Language Model Inference. *arXiv preprint arXiv:2603.29002*, 2026. URL <https://arxiv.org/abs/2603.29002>.
- [59] Amey Hengle, Prasoon Bajpai, Soham Dan, and Tanmoy Chakraborty. Multilingual Needle in a Haystack: Investigating Long-Context Behavior of Multilingual Large Language Models. *arXiv preprint arXiv:2408.10151*, 2024. URL <https://arxiv.org/abs/2408.10151>.
- [60] Coleman Hooper, Sehoon Kim, Hiva Mohammadzadeh, Monishwaran Maheswaran, Sebastian Zhao, June Paik, Michael W. Mahoney, Kurt Keutzer, and Amir Gholami. Squeezed Attention: Accelerating Long Context Length LLM Inference. *arXiv preprint arXiv:2411.09688*, 2024. URL <https://arxiv.org/abs/2411.09688>.
- [61] Yuki Hou, Haruki Tamoto, Qinghua Zhao, and Homei Miyashita. SynapticRAG: Enhancing Temporal Memory Retrieval in Large Language Models through Synaptic Mechanisms. *arXiv preprint arXiv:2410.13553*, 2024. URL <https://arxiv.org/abs/2410.13553>.
- [62] Cheng-Yu Hsieh, Yung-Sung Chuang, Chun-Liang Li, Zifeng Wang, Long T. Le, Abhishek Kumar,

- James Glass, Alexander Ratner, Chen-Yu Lee, Ranjay Krishna, and Tomas Pfister. Found in the Middle: Calibrating Positional Attention Bias Improves Long Context Utilization. *arXiv preprint arXiv:2406.16008*, 2024. URL <https://arxiv.org/abs/2406.16008>.
- [63] Chenxu Hu, Jie Fu, Chenzhuang Du, Simian Luo, Junbo Zhao, and Hang Zhao. ChatDB: Augmenting LLMs with Databases as Their Symbolic Memory. *arXiv preprint arXiv:2306.03901*, 2023. URL <https://arxiv.org/abs/2306.03901>.
- [64] Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, and Ping Luo. HiAgent: Hierarchical Working Memory Management for Solving Long-Horizon Agent Tasks with Large Language Model. *arXiv preprint arXiv:2408.09559*, 2024. URL <https://arxiv.org/abs/2408.09559>.
- [65] Wenbo Hu, Yining Hong, Yanjun Wang, Leison Gao, Zibu Wei, Xingcheng Yao, Nanyun Peng, Yonatan Bitton, Idan Szpektor, and Kai-Wei Chang. 3DLLM-Mem: Long-Term Spatial-Temporal Memory for Embodied 3D Large Language Model. *arXiv preprint arXiv:2505.22657*, 2025. URL <https://arxiv.org/abs/2505.22657>.
- [66] Yuanzhe Hu, Yu Wang, and Julian McAuley. Evaluating Memory in LLM Agents via Incremental Multi-Turn Interactions. *arXiv preprint arXiv:2507.05257*, 2025. URL <https://arxiv.org/abs/2507.05257>.
- [67] Zhiyuan Hu, Yuliang Liu, Jinman Zhao, Suyuchen Wang, Yan Wang, Wei Shen, Qing Gu, Anh Tuan Luu, See-Kiong Ng, Zhiwei Jiang, and Bryan Hooi. LongRecipe: Recipe for Efficient Long Context Generalization in Large Language Models. *arXiv preprint arXiv:2409.00509*, 2024. URL <https://arxiv.org/abs/2409.00509>.
- [68] Jiani Huang, Xingchen Zou, Lianghao Xia, and Qing Li. MR.Rec: Synergizing Memory and Reasoning for Personalized Recommendation Assistant with LLMs. *arXiv preprint arXiv:2510.14629*, 2025. URL <https://arxiv.org/abs/2510.14629>.
- [69] Mianqiu Huang, Xiaoran Liu, Shaojun Zhou, Mozhi Zhang, Qipeng Guo, Linyang Li, Chenkun Tan, Yang Gao, Pengyu Wang, Linlin Li, Qun Liu, Yaqian Zhou, Xipeng Qiu, and Xuanjing Huang. LongSafety: Enhance Safety for Long-Context LLMs. *arXiv preprint arXiv:2411.06899*, 2024. URL <https://arxiv.org/abs/2411.06899>.
- [70] Yunpeng Huang, Jingwei Xu, Junyu Lai, Zixu Jiang, Taolue Chen, Zenan Li, Yuan Yao, Xiaoxing Ma, Lijuan Yang, Hao Chen, Shupeng Li, and Penghao Zhao. Advancing Transformer Architecture in Long-Context Large Language Models: A Comprehensive Survey. *arXiv preprint arXiv:2311.12351*, 2023. URL <https://arxiv.org/abs/2311.12351>.
- [71] Yuxiang Huang, Binhang Yuan, Xu Han, Chaojun Xiao, and Zhiyuan Liu. Locret: Enhancing Eviction in Long-Context LLM Inference with Trained Retaining Heads on Consumer-Grade Devices. *arXiv preprint arXiv:2410.01805*, 2024. URL <https://arxiv.org/abs/2410.01805>.
- [72] Zhongzhan Huang, Guoming Ling, Shanshan Zhong, Hefeng Wu, and Liang Lin. MiniLongBench: The Low-cost Long Context Understanding Benchmark for Large Language Models. *arXiv preprint arXiv:2505.19959*, 2025. URL <https://arxiv.org/abs/2505.19959>.
- [73] Alexis Huet, Zied Ben Houidi, and Dario Rossi. Episodic Memories Generation and Evaluation Benchmark for Large Language Models. *arXiv preprint arXiv:2501.13121*, 2025. URL <https://arxiv.org/abs/2501.13121>.
- [74] Markus Huff and Elanur Ulakc. Towards a Psychology of Machines: Large Language Models Predict Human Memory. *arXiv preprint arXiv:2403.05152*, 2024. URL <https://arxiv.org/abs/2403.05152>.
- [75] Chenyu Hui. Towards robust long-context understanding of large language model via active recap learning. *arXiv preprint arXiv:2601.13734*, 2026. URL <https://arxiv.org/abs/2601.13734>.
- [76] Keito Inoshita. Who Does This Name Remind You of ? Nationality Prediction via Large Language Model Associative Memory. *arXiv preprint arXiv:2601.12771*, 2026. URL <https://arxiv.org/abs/2601.12771>.
- [77] Shuo Ji, Yibo Li, and Bryan Hooi. Memory is Reconstructed, Not Retrieved: Graph Memory for LLM Agents. *arXiv preprint arXiv:2606.06036*, 2026. URL <https://arxiv.org/abs/2606.06036>.
- [78] Suozhao Ji, Baodong Wu, Zehao Wang, Lei Xia, Qingping Li, Ruisong Wang, Wenbo Ding, Zhenhua Zhu, Boxun Li, Guohao Dai, and Yu Wang. Infini Memory: Maintainable Topic Documents for Long-Term LLM Agent Memory. *arXiv preprint arXiv:2606.10677*, 2026. URL <https://arxiv.org/abs/2606.10677>.
- [79] Junlong Jia, Xing Wu, Chaochen Gao, Ziyang Chen, Zijia Lin, Zhongzhi Li, Weinong Wang, Haotian Xu, Donghui Jin, Debing Zhang, and Binghui Guo. LiteLong: Resource-Efficient Long-Context Data Synthesis for LLMs. *arXiv preprint arXiv:2509.15568*, 2025. URL <https://arxiv.org/abs/2509.15568>.
- [80] Huiqiang Jiang, Qianhui Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt

- Compression. *arXiv preprint arXiv:2310.06839*, 2023. URL <https://arxiv.org/abs/2310.06839>.
- [81] Mingyu Jin, Weidi Luo, Sitao Cheng, Xinyi Wang, Wenyue Hua, Ruixiang Tang, William Yang Wang, and Yongfeng Zhang. Disentangling Memory and Reasoning Ability in Large Language Models. *arXiv preprint arXiv:2411.13504*, 2024. URL <https://arxiv.org/abs/2411.13504>.
- [82] Kiran Kate, Tejaswini Pedapati, Kinjal Basu, Yara Rizk, Vijil Chenthamarakshan, Subhajit Chaudhury, Mayank Agarwal, and Ibrahim Abdelaziz. LongFuncEval: Measuring the effectiveness of long context models for function calling. *arXiv preprint arXiv:2505.10570*, 2025. URL <https://arxiv.org/abs/2505.10570>.
- [83] Thomas Katraouras and Dimitrios Rafailidis. Memory Bank Compression for Continual Adaptation of Large Language Models. *arXiv preprint arXiv:2601.00756*, 2026. URL <https://arxiv.org/abs/2601.00756>.
- [84] Junwan Kim and Hyunkyung Bae. Reducing Peak Memory Usage for Modern Multimodal Large Language Model Pipelines. *arXiv preprint arXiv:2604.16734*, 2026. URL <https://arxiv.org/abs/2604.16734>.
- [85] Minjae Kim, Jinheon Baek, Soyeong Jeong, and Sung Ju Hwang. MemRefine: LLM-Guided Compression for Long-Term Agent Memory. *arXiv preprint arXiv:2606.13177*, 2026. URL <https://arxiv.org/abs/2606.13177>.
- [86] Wai-Chung Kwan, Kingshan Zeng, Yufei Wang, Yusen Sun, Liangyou Li, Lifeng Shang, Qun Liu, and Kam-Fai Wong. M4LE: A Multi-Ability Multi-Range Multi-Task Multi-Domain Long-Context Evaluation Benchmark for Large Language Models. *arXiv preprint arXiv:2310.19240*, 2023. URL <https://arxiv.org/abs/2310.19240>.
- [87] Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient Memory Management for Large Language Model Serving with PagedAttention. *arXiv preprint arXiv:2309.06180*, 2023. URL <https://arxiv.org/abs/2309.06180>.
- [88] Brandon Kynoch, Hugo Latapie, and Dwane van der Sluis. RecallM: An Adaptable Memory Mechanism with Temporal Understanding for Large Language Models. *arXiv preprint arXiv:2307.02738*, 2023. URL <https://arxiv.org/abs/2307.02738>.
- [89] Eric Lacosse, Mariana Duarte, Peter M. Todd, and Daniel C. McNamee. Emerging Human-like Strategies for Semantic Memory Foraging in Large Language Models. *arXiv preprint arXiv:2603.01822*, 2026. URL <https://arxiv.org/abs/2603.01822>.
- [90] Giuseppe Lando, Rosario Forte, Giovanni Maria Farinella, and Antonino Furnari. How Far Can Off-the-Shelf Multimodal Large Language Models Go in Online Episodic Memory Question Answering? *arXiv preprint arXiv:2506.16450*, 2025. URL <https://arxiv.org/abs/2506.16450>.
- [91] Hung Le and Svetha Venkatesh. Continual Fine-Tuning of Large Language Models via Program Memory. *arXiv preprint arXiv:2605.13162*, 2026. URL <https://arxiv.org/abs/2605.13162>.
- [92] Hung Le, Dai Do, Dung Nguyen, and Svetha Venkatesh. Reasoning Under 1 Billion: Memory-Augmented Reinforcement Learning for Large Language Models. *arXiv preprint arXiv:2504.02273*, 2025. URL <https://arxiv.org/abs/2504.02273>.
- [93] Changhun Lee, Minsang Seok, Jun gyu Jin, Younhyun Cho, and Eunhyeok Park. SEAL: Scaling to Emphasize Attention for Long-Context Retrieval. *arXiv preprint arXiv:2501.15225*, 2025. URL <https://arxiv.org/abs/2501.15225>.
- [94] Taewhoo Lee, Chanwoong Yoon, Kyochul Jang, Donghyeon Lee, Minju Song, Hyunjae Kim, and Jaewoo Kang. ETHIC: Evaluating Large Language Models on Long-Context Tasks with High Information Coverage. *arXiv preprint arXiv:2410.16848*, 2024. URL <https://arxiv.org/abs/2410.16848>.
- [95] Jingdi Lei, Di Zhang, Junxian Li, Weida Wang, Kaixuan Fan, Xiang Liu, Qihan Liu, Xiaoteng Ma, Baian Chen, and Soujanya Poria. \$\$-mem: Efficient Online Memory for Large Language Models. *arXiv preprint arXiv:2605.12357*, 2026. URL <https://arxiv.org/abs/2605.12357>.
- [96] Quinn Leng, Jacob Portes, Sam Havens, Matei Zaharia, and Michael Carbin. Long Context RAG Performance of Large Language Models. *arXiv preprint arXiv:2411.03538*, 2024. URL <https://arxiv.org/abs/2411.03538>.
- [97] Dongfang Li, Zetian Sun, Xinshuo Hu, Baotian Hu, and Min Zhang. CMT: A Memory Compression Method for Continual Knowledge Learning of Large Language Models. *arXiv preprint arXiv:2412.07393*, 2024. URL <https://arxiv.org/abs/2412.07393>.
- [98] Jiaqi Li, Mengmeng Wang, Zilong Zheng, and Muhan Zhang. LooGLE: Can Long-Context Language Models Understand Long Contexts? *arXiv preprint arXiv:2311.04939*, 2023. URL <https://arxiv.org/abs/2311.04939>.
- [99] Jiayi Li, Xingxing Zhang, Xun Wang, Xiaolong Huang, Li Dong, Liang Wang, Si-Qing Chen, Wei Lu, and Furu Wei. WildLong: Synthesizing Realistic Long-Context Instruction Data at Scale. *arXiv preprint arXiv:2502.16684*, 2025. URL <https://arxiv.org/abs/2502.16684>.

- [100] Junsong Li, Jie Zhou, Bihao Zhan, Yutao Yang, Qianjun Pan, Shilian Chen, Tianyu Huai, Xin Li, Qin Chen, and Liang He. LifeAlign: Life-long Alignment for Large Language Models with Memory-Augmented Focalized Preference Optimization. *arXiv preprint arXiv:2509.17183*, 2025. URL <https://arxiv.org/abs/2509.17183>.
- [101] Kangcong Li, Peng Ye, Chongjun Tu, Lin Zhang, Chunfeng Song, Jiamin Wu, Tao Yang, Qihao Zheng, and Tao Chen. PaceLLM: Brain-Inspired Large Language Models for Long-Context Understanding. *arXiv preprint arXiv:2506.17310*, 2025. URL <https://arxiv.org/abs/2506.17310>.
- [102] Miao Li, Irina Saparina, Alexander Gurung, and Mirella Lapata. Long-Context Reasoning Through Proxy-Based Chain-of-Thought Tuning. *arXiv preprint arXiv:2605.20201*, 2026. URL <https://arxiv.org/abs/2605.20201>.
- [103] Shengzhi Li, Kittipat Kampa, Rongyu Lin, Bohang Li, and Shichao Pei. Abstract2Appendix: Academic Reviews Enhance LLM Long-Context Capabilities. *arXiv preprint arXiv:2411.05232*, 2024. URL <https://arxiv.org/abs/2411.05232>.
- [104] Shilong Li, Yancheng He, Hangyu Guo, Xingyuan Bu, Ge Bai, Jie Liu, Jiaheng Liu, Xingwei Qu, Yangguang Li, Wanli Ouyang, Wenbo Su, and Bo Zheng. GraphReader: Building Graph-based Agent to Enhance Long-Context Abilities of Large Language Models. *arXiv preprint arXiv:2406.14550*, 2024. URL <https://arxiv.org/abs/2406.14550>.
- [105] Siheng Li, Cheng Yang, Zesen Cheng, Lemao Liu, Mo Yu, Yujiu Yang, and Wai Lam. Large Language Models Can Self-Improve in Long-context Reasoning. *arXiv preprint arXiv:2411.08147*, 2024. URL <https://arxiv.org/abs/2411.08147>.
- [106] Tianle Li, Ge Zhang, Quy Duc Do, Xiang Yue, and Wenhui Chen. Long-context LLMs Struggle with Long In-context Learning. *arXiv preprint arXiv:2404.02060*, 2024. URL <https://arxiv.org/abs/2404.02060>.
- [107] Weizhuo Li, Zhigang Wang, Yu Gu, and Ge Yu. XKV: Personalized KV Cache Memory Reduction for Long-Context LLM Inference. *arXiv preprint arXiv:2412.05896*, 2024. URL <https://arxiv.org/abs/2412.05896>.
- [108] Wenxuan Li, Zhenfei Zhang, Mi Zhang, Geng Hong, Mi Wen, Xiaoyu You, and Min Yang. From Anchors to Supervision: Memory-Graph Guided Corpus-Free Unlearning for Large Language Models. *arXiv preprint arXiv:2604.13777*, 2026. URL <https://arxiv.org/abs/2604.13777>.
- [109] Yuepei Li, Kang Zhou, Qiao Qiao, Bach Nguyen, Qing Wang, and Qi Li. Investigating Context-Faithfulness in Large Language Models: The Roles of Memory Strength and Evidence Style. *arXiv preprint arXiv:2409.10955*, 2024. URL <https://arxiv.org/abs/2409.10955>.
- [110] Zhiyu Li, Shichao Song, Hanyu Wang, Simin Niu, Ding Chen, Jiawei Yang, Chenyang Xi, Huayi Lai, Jihao Zhao, Yezhaohui Wang, Junpeng Ren, Zehao Lin, Jiahao Huo, Tianyi Chen, Kai Chen, Kehang Li, Zhiqiang Yin, Qingchen Yu, Bo Tang, Hongkang Yang, et al. MemOS: An Operating System for Memory-Augmented Generation (MAG) in Large Language Models. *arXiv preprint arXiv:2505.22101*, 2025. URL <https://arxiv.org/abs/2505.22101>.
- [111] Haoran Lian, Junmin Chen, Wei Huang, Yizhe Xiong, Wenping Hu, Guiguang Ding, Hui Chen, Jianwei Niu, Zijia Lin, Fuzheng Zhang, and Di Zhang. Breaking the Stage Barrier: A Novel Single-Stage Approach to Long Context Extension for Large Language Models. *arXiv preprint arXiv:2412.07171*, 2024. URL <https://arxiv.org/abs/2412.07171>.
- [112] Manlai Liang, Wanyi Huang, Mandi Liu, Huaijun Li, and Jinlong Li. Lag-Relative Sparse Attention In Long Context Training. *arXiv preprint arXiv:2506.11498*, 2025. URL <https://arxiv.org/abs/2506.11498>.
- [113] Zihan Liao, Jun Wang, Hang Yu, Lingxiao Wei, Jianguo Li, Jun Wang, and Wei Zhang. E2LLM: Encoder Elongated Large Language Models for Long-Context Understanding and Reasoning. *arXiv preprint arXiv:2409.06679*, 2024. URL <https://arxiv.org/abs/2409.06679>.
- [114] Yong-Cheng Liaw and Shuo-Han Chen. MemAscend: System Memory Optimization for SSD-Offloaded LLM Fine-Tuning. *arXiv preprint arXiv:2505.23254*, 2025. URL <https://arxiv.org/abs/2505.23254>.
- [115] Jingyang Lin, Andy Wong, Tian Xia, Shenghua He, Hui Wei, Mei Han, and Jiebo Luo. Facilitating Long Context Understanding via Supervised Chain-of-Thought Reasoning. *arXiv preprint arXiv:2502.13127*, 2025. URL <https://arxiv.org/abs/2502.13127>.
- [116] Zhan Ling, Kang Liu, Kai Yan, Yifan Yang, Weijian Lin, Ting-Han Fan, Lingfeng Shen, Zhengyin Du, and Jiecao Chen. LongReason: A Synthetic Long-Context Reasoning Benchmark via Context Expansion. *arXiv preprint arXiv:2501.15089*, 2025. URL <https://arxiv.org/abs/2501.15089>.
- [117] Jiaheng Liu, Dawei Zhu, Zhiqi Bai, Yancheng He, Huanxuan Liao, Haoran Que, Zekun Wang, Chenchen Zhang, Ge Zhang, Jiebin Zhang, Yuanxing Zhang, Zhuo Chen, Hangyu Guo, Shilong Li, Ziqiang Liu, Yong Shan, Yifan Song, Jiayi Tian, Wenhao Wu, Zhejian Zhou, et al. A Compre-

- hensive Survey on Long Context Language Modeling. *arXiv preprint arXiv:2503.17407*, 2025. URL <https://arxiv.org/abs/2503.17407>.
- [118] Jiawei Liu, Jia Le Tian, Vijay Daita, Yuxiang Wei, Yifeng Ding, Yuhang Katherine Wang, Jun Yang, and Lingming Zhang. RepoQA: Evaluating Long Context Code Understanding. *arXiv preprint arXiv:2406.06025*, 2024. URL <https://arxiv.org/abs/2406.06025>.
- [119] Junhao Liu, Haonan Yu, Zhenyu Yan, and Xin Zhang. Focus-LIME: Surgical Interpretation of Long-Context Large Language Models via Proxy-Based Neighborhood Selection. *arXiv preprint arXiv:2602.04607*, 2026. URL <https://arxiv.org/abs/2602.04607>.
- [120] Lei Liu, Xiaoyan Yang, Yue Shen, Binbin Hu, Zhiqiang Zhang, Jinjie Gu, and Guannan Zhang. Think-in-Memory: Recalling and Post-thinking Enable LLMs with Long-Term Memory. *arXiv preprint arXiv:2311.08719*, 2023. URL <https://arxiv.org/abs/2311.08719>.
- [121] Lian Liu, Shixin Zhao, Yutian Zhou, Yintao He, Mengdi Wang, Yinhe Han, and Ying Wang. PAM: Processing Across Memory Hierarchy for Efficient KV-centric LLM Serving System. *arXiv preprint arXiv:2602.11521*, 2026. URL <https://arxiv.org/abs/2602.11521>.
- [122] Na Liu, Liangyu Chen, Xiaoyu Tian, Wei Zou, Kaijiang Chen, and Ming Cui. From LLM to Conversational Agent: A Memory Enhanced Architecture with Fine-Tuning of Large Language Models. *arXiv preprint arXiv:2401.02777*, 2024. URL <https://arxiv.org/abs/2401.02777>.
- [123] Ruitong Liu, Boxu Lin, Peize Li, Siyuan Li, Yunjia Wu, Te Sun, and Chaohan Wu. Beyond Prefixes: Graph-as-Memory Cross-Attention for Knowledge Graph Completion with Large Language Models. *arXiv preprint arXiv:2510.08966*, 2025. URL <https://arxiv.org/abs/2510.08966>.
- [124] Siyi Liu, Kishalay Halder, Zheng Qi, Wei Xiao, Nikolaos Pappas, Phu Mon Htut, Neha Anna John, Yassine Benajiba, and Dan Roth. Towards Long Context Hallucination Detection. *arXiv preprint arXiv:2504.19457*, 2025. URL <https://arxiv.org/abs/2504.19457>.
- [125] Weichu Liu, Jing Xiong, Yuxuan Hu, Zixuan Li, Minghuan Tan, Ningning Mao, Hui Shen, Wendong Xu, Chaofan Tao, Min Yang, Chengming Li, Lingpeng Kong, and Ngai Wong. LongEmotion: Measuring Emotional Intelligence of Large Language Models in Long-Context Interaction. *arXiv preprint arXiv:2509.07403*, 2025. URL <https://arxiv.org/abs/2509.07403>.
- [126] Weihao Liu, Ning Wu, Shiping Yang, Wenbiao Ding, Shining Liang, Ming Gong, and Dongmei Zhang. MuDAF: Long-Context Multi-Document Attention Focusing through Contrastive Learning on Attention Heads. *arXiv preprint arXiv:2502.13963*, 2025. URL <https://arxiv.org/abs/2502.13963>.
- [127] WenTao Liu, Ruohua Zhang, Aimin Zhou, Feng Gao, and JiaLi Liu. Echo: A Large Language Model with Temporal Episodic Memory. *arXiv preprint arXiv:2502.16090*, 2025. URL <https://arxiv.org/abs/2502.16090>.
- [128] Xiang Liu, Peijie Dong, Xuming Hu, and Xiaowen Chu. LongGenBench: Long-context Generation Benchmark. *arXiv preprint arXiv:2410.04199*, 2024. URL <https://arxiv.org/abs/2410.04199>.
- [129] Xiaoran Liu, Ruixiao Li, Mianqiu Huang, Zhigeng Liu, Yuerong Song, Qipeng Guo, Siyang He, Qiqi Wang, Linlin Li, Qun Liu, Ziwei He, Yaqian Zhou, Xuanjing Huang, and Xipeng Qiu. Thus Spake Long-Context Large Language Model. *arXiv preprint arXiv:2502.17129*, 2025. URL <https://arxiv.org/abs/2502.17129>.
- [130] Lingkun Long, Rubing Yang, Yushi Huang, Desheng Hui, Ao Zhou, and Jianlei Yang. SlimInfer: Accelerating Long-Context LLM Inference via Dynamic Token Pruning. *arXiv preprint arXiv:2508.06447*, 2025. URL <https://arxiv.org/abs/2508.06447>.
- [131] Lingkun Long, Yushi Huang, Shihao Bai, Ruihao Gong, Jun Zhang, Ao Zhou, and Jianlei Yang. Focus-dLLM: Accelerating Long-Context Diffusion LLM Inference via Confidence-Guided Context Focusing. *arXiv preprint arXiv:2602.02159*, 2026. URL <https://arxiv.org/abs/2602.02159>.
- [132] Yuyang Lou and Charles Li. Dynamic Long Short-Term Memory Based Memory Storage For Long Horizon LLM Interaction. *arXiv preprint arXiv:2507.03042*, 2025. URL <https://arxiv.org/abs/2507.03042>.
- [133] Yedidel Louck. Securing LLM-Agent Long-Term Memory Against Poisoning: Non-Malleable, Origin-Bound Authority with Machine-Checked Guarantees. *arXiv preprint arXiv:2606.24322*, 2026. URL <https://arxiv.org/abs/2606.24322>.
- [134] Mingfei Lu, Mengjia Wu, Feng Liu, Jiawei Xu, Weikai Li, Haoyang Wang, Zhengdong Hu, Ying Ding, Yizhou Sun, Jie Lu, and Yi Zhang. Choosing How to Remember: Adaptive Memory Structures for LLM Agents. *arXiv preprint arXiv:2602.14038*, 2026. URL <https://arxiv.org/abs/2602.14038>.
- [135] Yida Lu, Jiale Cheng, Zhexin Zhang, Shiyao Cui, Cunxiang Wang, Xiaotao Gu, Yuxiao Dong, Jie Tang, Hongning Wang, and Minlie Huang. LongSafety: Evaluating Long-Context Safety of Large Language Models. *arXiv preprint*

- arXiv:2502.16971*, 2025. URL <https://arxiv.org/abs/2502.16971>.
- [136] Kun Luo, Zheng Liu, Shitao Xiao, and Kang Liu. BGE Landmark Embedding: A Chunking-Free Embedding Method For Retrieval Augmented Long-Context Large Language Models. *arXiv preprint arXiv:2402.11573*, 2024. URL <https://arxiv.org/abs/2402.11573>.
- [137] Kun Luo, Zheng Liu, Peitian Zhang, Hongjin Qian, Jun Zhao, and Kang Liu. Does RAG Really Perform Bad For Long-Context Processing? *arXiv preprint arXiv:2502.11444*, 2025. URL <https://arxiv.org/abs/2502.11444>.
- [138] Qijun Luo, Hengxu Yu, and Xiao Li. BAdam: A Memory Efficient Full Parameter Optimization Method for Large Language Models. *arXiv preprint arXiv:2404.02827*, 2024. URL <https://arxiv.org/abs/2404.02827>.
- [139] Chenxiang Ma, Zhisheng Ye, Hanyu Zhao, Zehua Yang, Tianhao Fu, Jiaxun Han, Jie Zhang, Yingwei Luo, Xiaolin Wang, Zhenlin Wang, Yong Li, and Diyu Zhou. Memory Offloading for Large Language Model Inference with Latency SLO Guarantees. *arXiv preprint arXiv:2502.08182*, 2025. URL <https://arxiv.org/abs/2502.08182>.
- [140] Wenquan Ma, Jiayan Nan, Wenlong Wu, and Yize Chen. What Deserves Memory: Adaptive Memory Distillation for LLM Agents. *arXiv preprint arXiv:2508.03341*, 2025. URL <https://arxiv.org/abs/2508.03341>.
- [141] Kishan Maharaj, Nandakishore Menon, Ashita Saxena, and Srikanth Tamilselvam. Robustness and Reasoning Fidelity of Large Language Models in Long-Context Code Question Answering. *arXiv preprint arXiv:2602.17183*, 2026. URL <https://arxiv.org/abs/2602.17183>.
- [142] Jinjie Mai, Jun Chen, Bing Li, Guocheng Qian, Mohamed Elhoseiny, and Bernard Ghanem. LLM as A Robotic Brain: Unifying Egocentric Memory and Control. *arXiv preprint arXiv:2304.09349*, 2023. URL <https://arxiv.org/abs/2304.09349>.
- [143] Yansheng Mao, Jiaqi Li, Fanxu Meng, Jing Xiong, Zilong Zheng, and Muhan Zhang. LIFT: Improving Long Context Understanding Through Long Input Fine-Tuning. *arXiv preprint arXiv:2412.13626*, 2024. URL <https://arxiv.org/abs/2412.13626>.
- [144] Fathima Mashood and Mohamed Nabeel. Can LLMs Reason About Brand Ownership? An Empirical Study of Domain Attribution Intelligence. *arXiv preprint arXiv:2606.20868*, 2026. URL <https://arxiv.org/abs/2606.20868>.
- [145] Gregory Matsnev. Uncertainty Decomposition for Clarification Seeking in LLM Agents. *arXiv preprint arXiv:2606.19559*, 2026. URL <https://arxiv.org/abs/2606.19559>.
- [146] Avni Mittal. Did You Forget What I Asked? Prospective Memory Failures in Large Language Models. *arXiv preprint arXiv:2603.23530*, 2026. URL <https://arxiv.org/abs/2603.23530>.
- [147] Pearl Mody, Mihir Panchal, Rishit Kar, Kiran Bhowmick, and Ruhina Karani. CraniMem: Cranial Inspired Gated and Bounded Memory for Agentic Systems. *arXiv preprint arXiv:2603.15642*, 2026. URL <https://arxiv.org/abs/2603.15642>.
- [148] Elliot Nelson, Georgios Kollias, Payel Das, Subhajit Chaudhury, and Soham Dan. Needle in the Haystack for Memory Based Large Language Models. *arXiv preprint arXiv:2407.01437*, 2024. URL <https://arxiv.org/abs/2407.01437>.
- [149] Narjes Nourzad and Carlee Joe-Wong. MIRA: Memory-Integrated Reinforcement Learning Agent with Limited LLM Guidance. *arXiv preprint arXiv:2602.17930*, 2026. URL <https://arxiv.org/abs/2602.17930>.
- [150] A. Emin Orhan. Recognition, recall, and retention of few-shot memories in large language models. *arXiv preprint arXiv:2303.17557*, 2023. URL <https://arxiv.org/abs/2303.17557>.
- [151] Aadi Palnitkar, Mingyang Mao, Nicholas Waytowich, Vinicius G. Goecks, and Xiaomin Lin. MILSCORE: Benchmarking Long-Context Geospatial Reasoning and Planning in Large Language Models. *arXiv preprint arXiv:2601.21826*, 2026. URL <https://arxiv.org/abs/2601.21826>.
- [152] Junyoung Park, Seongyong Ju, Sunghwan Park, and Jaewoo Lee. Persona Attack: Incremental Memory Injection Jailbreak Attack against Large Language Models. *arXiv preprint arXiv:2606.00150*, 2026. URL <https://arxiv.org/abs/2606.00150>.
- [153] Yeonhong Park, Jake Hyun, SangLyul Cho, Bonggeun Sim, and Jae W. Lee. Any-Precision LLM: Low-Cost Deployment of Multiple, Different-Sized LLMs. *arXiv preprint arXiv:2402.10517*, 2024. URL <https://arxiv.org/abs/2402.10517>.
- [154] Mitchell Piehl, Zhaohan Xi, Zuobin Xiong, Pan He, and Muchao Ye. ER-MIA: Black-Box Adversarial Memory Injection Attacks on Long-Term Memory-Augmented Large Language Models. *arXiv preprint arXiv:2602.15344*, 2026. URL <https://arxiv.org/abs/2602.15344>.
- [155] Mathis Pink, Vy A. Vo, Qinyuan Wu, Jianing Mu, Javier S. Turek, Uri Hasson, Kenneth A. Norman, Sebastian Michelmann, Alexander Huth, and Mariya Toneva. Assessing Episodic Memory in LLMs with Sequence Order Recall Tasks. *arXiv*

- preprint *arXiv:2410.08133*, 2024. URL <https://arxiv.org/abs/2410.08133>.
- [156] Mathis Pink, Qinyuan Wu, Vy Ai Vo, Javier Turek, Jianing Mu, Alexander Huth, and Mariya Toneva. Position: Episodic Memory is the Missing Piece for Long-Term LLM Agents. *arXiv preprint arXiv:2502.06975*, 2025. URL <https://arxiv.org/abs/2502.06975>.
 - [157] Ramya Prabhu, Ajay Nayak, Jayashree Mohan, Ramachandran Ramjee, and Ashish Panwar. vAttention: Dynamic Memory Management for Serving LLMs without PagedAttention. *arXiv preprint arXiv:2405.04437*, 2024. URL <https://arxiv.org/abs/2405.04437>.
 - [158] Sidharth Pulipaka, Stanislaw Hlebig, Leonidas Raghav, Sahar Abdelnabi, Vyas Raina, Ivaxi Sheth, and Mario Fritz. Hidden in Memory: Sleeper Memory Poisoning in LLM Agents. *arXiv preprint arXiv:2605.15338*, 2026. URL <https://arxiv.org/abs/2605.15338>.
 - [159] Hongjin Qian, Zheng Liu, Peitian Zhang, Kelong Mao, Defu Lian, Zhicheng Dou, and Tiejun Huang. MemoRAG: Boosting Long Context Processing with Global Memory-Enhanced Retrieval Augmentation. *arXiv preprint arXiv:2409.05591*, 2024. URL <https://arxiv.org/abs/2409.05591>.
 - [160] Jielin Qiu, Zuxin Liu, Zhiwei Liu, Rithesh Murthy, Jianguo Zhang, Haolin Chen, Shiyu Wang, Ming Zhu, Liangwei Yang, Juntao Tan, Zhepeng Cen, Cheng Qian, Shelby Heinecke, Weiran Yao, Silvio Savarese, Caiming Xiong, and Huan Wang. LoCoBench: A Benchmark for Long-Context Large Language Models in Complex Software Engineering. *arXiv preprint arXiv:2509.09614*, 2025. URL <https://arxiv.org/abs/2509.09614>.
 - [161] Zexuan Qiu, Jingjing Li, Shijue Huang, Xiaoli Jiao, Wanjuan Zhong, and Irwin King. CLongEval: A Chinese Benchmark for Evaluating Long-Context Large Language Models. *arXiv preprint arXiv:2403.03514*, 2024. URL <https://arxiv.org/abs/2403.03514>.
 - [162] Ryan Wei Heng Quek, Sanghyuk Lee, Alfred Wei Lun Leong, Arun Verma, Alok Prakash, Nancy F. Chen, Bryan Kian Hsiang Low, Daniela Rus, and Armando Solar-Lezama. MeMo: Memory as a Model. *arXiv preprint arXiv:2605.15156*, 2026. URL <https://arxiv.org/abs/2605.15156>.
 - [163] Shreyas Rajesh, Pavan Holur, Chenda Duan, David Chong, and Vwani Roychowdhury. Beyond Fact Retrieval: Episodic Memory for RAG with Generative Semantic Workspaces. *arXiv preprint arXiv:2511.07587*, 2025. URL <https://arxiv.org/abs/2511.07587>.
 - [164] Shreyas Rajesh, Pavan Holur, Mehmet Yigit Turali, Chenda Duan, and Vwani Roychowdhury. Panini: Continual Learning in Token Space via Structured Memory. *arXiv preprint arXiv:2602.15156*, 2026. URL <https://arxiv.org/abs/2602.15156>.
 - [165] Miguel Moura Ramos, Duarte M. Alves, and Andre F. T. Martins. A Recipe for Long-Context Reasoning in Large Language Models via On-Policy Optimization and Distillation. *arXiv preprint arXiv:2605.12227*, 2026. URL <https://arxiv.org/abs/2605.12227>.
 - [166] Zixin Rao, Wentian Zhu, Chan Aristella Lu, Zhaorun Chen, Wei Niu, Le Guan, Bo Li, and Zhen Xiang. FragFuse: Bypassing Access Control of Large Language Model Agents via Memory-Based Query Fragmentation and Fusion. *arXiv preprint arXiv:2606.15609*, 2026. URL <https://arxiv.org/abs/2606.15609>.
 - [167] Pranath Reddy. Selective Memory Retention for Long-Horizon LLM Agents. *arXiv preprint arXiv:2606.29178*, 2026. URL <https://arxiv.org/abs/2606.29178>.
 - [168] Xubin Ren, Lingrui Xu, Long Xia, Shuaiqiang Wang, Dawei Yin, and Chao Huang. VideoRAG: Retrieval-Augmented Generation with Extreme Long-Context Videos. *arXiv preprint arXiv:2502.01549*, 2025. URL <https://arxiv.org/abs/2502.01549>.
 - [169] Shahnewaz Karim Sakib and Anindya Bijoy Das. Memory as an Attack Surface in LLM Agents: A Study on Multiple-Choice Question Answering. *arXiv preprint arXiv:2606.29030*, 2026. URL <https://arxiv.org/abs/2606.29030>.
 - [170] Rana Salama, Jason Cai, Michelle Yuan, Anna Currey, Monica Sunkara, Yi Zhang, and Yassine Benajiba. MemInsight: Autonomous Memory Augmentation for LLM Agents. *arXiv preprint arXiv:2503.21760*, 2025. URL <https://arxiv.org/abs/2503.21760>.
 - [171] Cosmo Santoni. Contextual Memory Virtualisation: DAG-Based State Management and Structurally Lossless Trimming for LLM Agents. *arXiv preprint arXiv:2602.22402*, 2026. URL <https://arxiv.org/abs/2602.22402>.
 - [172] Ali Sarabadani and Mahtab Tajvidiyan. DYNA: Dynamic Episodic Memory Networks for Augmenting Large Language Models with Temporal Knowledge Graphs in Continuous Learning. *arXiv preprint arXiv:2606.15778*, 2026. URL <https://arxiv.org/abs/2606.15778>.
 - [173] Gabriel Sarch, Yue Wu, Michael J. Tarr, and Katerina Fragkiadaki. Open-Ended Instructable Embodied Agents with Memory-Augmented Large Language Models. *arXiv preprint arXiv:2310.15127*, 2023. URL <https://arxiv.org/abs/2310.15127>.

- [174] Ignacio Sastre and Aiala Rosa. Memory Tokens: Large Language Models Can Generate Reversible Sentence Embeddings. *arXiv preprint arXiv:2506.15001*, 2025. URL <https://arxiv.org/abs/2506.15001>.
- [175] Simon Schug and Brenden M. Lake. Are they human? Detecting large language models by probing human memory constraints. *arXiv preprint arXiv:2604.00016*, 2026. URL <https://arxiv.org/abs/2604.00016>.
- [176] Lianlei Shan, Shixian Luo, Zezhou Zhu, Yu Yuan, and Yong Wu. Cognitive Memory in Large Language Models. *arXiv preprint arXiv:2504.02441*, 2025. URL <https://arxiv.org/abs/2504.02441>.
- [177] Xianpeng Shang, Jiang Li, Zehua Duo, Qianyi Cai, and Xiangdong Su. Training-Inference Consistent Segmented Execution for Long-Context LLMs. *arXiv preprint arXiv:2605.11744*, 2026. URL <https://arxiv.org/abs/2605.11744>.
- [178] Wei Shen, Zhang Yaxiang, Minhui Huang, Mengfan Xu, Jiawei Zhang, and Cong Shen. MLorC: Momentum Low-rank Compression for Memory Efficient Large Language Model Adaptation. *arXiv preprint arXiv:2506.01897*, 2025. URL <https://arxiv.org/abs/2506.01897>.
- [179] Zhanyu Shen, Sijie Cheng, Zhicheng Guo, Weiqin Wang, Yile Wang, and Hui Huang. AnchorMem: Anchored Facts with Associative Contexts for Building Memory in Large Language Models. *arXiv preprint arXiv:2604.17377*, 2026. URL <https://arxiv.org/abs/2604.17377>.
- [180] Boheng Sheng, Jiacheng Yao, Meicong Zhang, and Guoxiu He. Dynamic Chunking and Selection for Reading Comprehension of Ultra-Long Context in Large Language Models. *arXiv preprint arXiv:2506.00773*, 2025. URL <https://arxiv.org/abs/2506.00773>.
- [181] Yaorui Shi, Yuxin Chen, Siyuan Wang, Sihang Li, Hengxing Cai, Qi Gu, Xiang Wang, and An Zhang. Look Back to Reason Forward: Revisitable Memory for Long-Context LLM Agents. *arXiv preprint arXiv:2509.23040*, 2025. URL <https://arxiv.org/abs/2509.23040>.
- [182] Igor Shilov, Matthieu Meeus, and Yves-Alexandre de Montjoye. The Mosaic Memory of Large Language Models. *arXiv preprint arXiv:2405.15523*, 2024. URL <https://arxiv.org/abs/2405.15523>.
- [183] Saish Sachin Shinde. SCM: Sleep-Consolidated Memory with Algorithmic Forgetting for Large Language Models. *arXiv preprint arXiv:2604.20943*, 2026. URL <https://arxiv.org/abs/2604.20943>.
- [184] Yakov Pyotr Shkolnikov. Agent Memory Below the Prompt: Persistent Q4 KV Cache for Multi-Agent LLM Inference on Edge Devices. *arXiv preprint arXiv:2603.04428*, 2026. URL <https://arxiv.org/abs/2603.04428>.
- [185] Alina Shutova, Alexandra Olenina, Ivan Vinogradov, and Anton Sinitin. Evaluating Memory Structure in LLM Agents. *arXiv preprint arXiv:2602.11243*, 2026. URL <https://arxiv.org/abs/2602.11243>.
- [186] Shuzheng Si, Haozhe Zhao, Gang Chen, Yunshui Li, Kangyang Luo, Chuancheng Lv, Kaikai An, Fanchao Qi, Baobao Chang, and Maosong Sun. GATEAU: Selecting Influential Samples for Long Context Alignment. *arXiv preprint arXiv:2410.15633*, 2024. URL <https://arxiv.org/abs/2410.15633>.
- [187] Dingjie Song, Shunian Chen, Guiming Hardy Chen, Fei Yu, Xiang Wan, and Benyou Wang. MileBench: Benchmarking MLLMs in Long Context. *arXiv preprint arXiv:2404.18532*, 2024. URL <https://arxiv.org/abs/2404.18532>.
- [188] Mingyang Song, Mao Zheng, and Xuan Luo. Counting-Stars: A Multi-evidence, Position-aware, and Scalable Benchmark for Evaluating Long-Context Large Language Models. *arXiv preprint arXiv:2403.11802*, 2024. URL <https://arxiv.org/abs/2403.11802>.
- [189] Vasundra Srinivasan. Stateless Decision Memory for Enterprise AI Agents. *arXiv preprint arXiv:2604.20158*, 2026. URL <https://arxiv.org/abs/2604.20158>.
- [190] Konrad Staniszewski, Szymon Tworowski, Sebastian Jaszczur, Yu Zhao, Henryk Michalewski, ukasz Kucinski, and Piotr Mios. Structured Packing in LLM Training Improves Long Context Utilization. *arXiv preprint arXiv:2312.17296*, 2023. URL <https://arxiv.org/abs/2312.17296>.
- [191] Segaran Subramanian and Abhigya Verma. Modular Techniques for Synthetic Long-Context Data Generation in Language Model Training and Evaluation. *arXiv preprint arXiv:2509.01185*, 2025. URL <https://arxiv.org/abs/2509.01185>.
- [192] Haoran Sun and Shaoning Zeng. Hierarchical Memory for High-Efficiency Long-Term Reasoning in LLM Agents. *arXiv preprint arXiv:2507.22925*, 2025. URL <https://arxiv.org/abs/2507.22925>.
- [193] Haoran Sun, Zekun Zhang, and Shaoning Zeng. Preference-Aware Memory Update for Long-Term LLM Agents. *arXiv preprint arXiv:2510.09720*, 2025. URL <https://arxiv.org/abs/2510.09720>.
- [194] Jiejun Tan, Zhicheng Dou, Liancheng Zhang, Yuyang Hu, Yiruo Cheng, and Ji-Rong Wen. MemSifter: Offloading LLM Memory Retrieval via Outcome-Driven Proxy Reasoning. *arXiv preprint*

- arXiv:2603.03379*, 2026. URL <https://arxiv.org/abs/2603.03379>.
- [195] Sijun Tan, Xiuyu Li, Shishir Patil, Ziyang Wu, Tianjun Zhang, Kurt Keutzer, Joseph E. Gonzalez, and Raluca Ada Popa. LLoCO: Learning Long Contexts Offline. *arXiv preprint arXiv:2404.07979*, 2024. URL <https://arxiv.org/abs/2404.07979>.
- [196] Xingyu Tan, Xiaoyang Wang, Qing Liu, Xiwei Xu, Xin Yuan, Liming Zhu, and Wenjie Zhang. MemoTime: Memory-Augmented Temporal Knowledge Graph Enhanced Large Language Model Reasoning. *arXiv preprint arXiv:2510.13614*, 2025. URL <https://arxiv.org/abs/2510.13614>.
- [197] Jiwei Tang, Jin Xu, Tingwei Lu, Zhicheng Zhang, Yiming Zhao, Lin Hai, and Hai-Tao Zheng. Perception Compressor: A Training-Free Prompt Compression Framework in Long Context Scenarios. *arXiv preprint arXiv:2409.19272*, 2024. URL <https://arxiv.org/abs/2409.19272>.
- [198] Zecheng Tang, Haitian Wang, Quantong Qiu, Baibei Ji, Ruoxi Sun, Keyan Zhou, Juntao Li, and Min Zhang. LOOM-Scope: a comprehensive and efficient LLong-cOntext Model evaluation framework. *arXiv preprint arXiv:2507.04723*, 2025. URL <https://arxiv.org/abs/2507.04723>.
- [199] Zecheng Tang, Baibei Ji, Ruoxi Sun, Haitian Wang, Wangjie You, Zhang Yijun, Wenpeng Zhu, Ji Qi, Juntao Li, and Min Zhang. MemoryRewardBench: Benchmarking Reward Models for Long-Term Memory Management in Large Language Models. *arXiv preprint arXiv:2601.11969*, 2026. URL <https://arxiv.org/abs/2601.11969>.
- [200] PSBC LLM Team, Huawei LLM Team, Ruihan Long, Junjie Wu, Tianan Zhang, Duo Zhang, Yaozong Wu, Jinbin Fu, Chang Liu, Zhentao Tang, Wenshuang Yang, Xin Wang, Zhihao Song, Ning Huang, Wenjing Xu, Shuai Zong, Shupe Sun, Sen Wang, Jing Hu, Bin Wang, et al. YouZhi: Towards High-Concurrency Financial LLMs via Adaptive GQA-to-MLA Transition. *arXiv preprint arXiv:2606.05868*, 2026. URL <https://arxiv.org/abs/2606.05868>.
- [201] Utkarsh Tiwari, Aryan Seth, Adi Mukherjee, Kaavya Mer, Kavish, and Dhruv Kumar. DebateBench: A Challenging Long Context Reasoning Benchmark For Large Language Models. *arXiv preprint arXiv:2502.06279*, 2025. URL <https://arxiv.org/abs/2502.06279>.
- [202] Aditya Tomar, Coleman Hooper, Minjae Lee, Haocheng Xi, Rishabh Tiwari, Wonjun Kang, Luca Manolache, Michael W. Mahoney, Kurt Keutzer, and Amir Gholami. XQuant: Breaking the Memory Wall for LLM Inference with KV Cache Rematerialization. *arXiv preprint arXiv:2508.10395*, 2025. URL <https://arxiv.org/abs/2508.10395>.
- [203] Shubhangi Upasani, Ravi Shanker Raju, Bo Li, Mengmeng Ji, John Long, Chen Wu, Urmish Thakker, and Guangtao Wang. Cross-Family Speculative Prefill: Training-Free Long-Context Compression with Small Draft Models. *arXiv preprint arXiv:2603.02631*, 2026. URL <https://arxiv.org/abs/2603.02631>.
- [204] Harshil Vejendla. H1B-KV: Hybrid One-Bit Caches for Memory-Efficient Large Language Model Inference. *arXiv preprint arXiv:2510.05529*, 2025. URL <https://arxiv.org/abs/2510.05529>.
- [205] Kiran Vodrahalli, Santiago Ontanon, Nilesh Tripuraneni, Kelvin Xu, Sanil Jain, Rakesh Shivanna, Jeffrey Hui, Nishanth Dikkala, Mehran Kazemi, Bahare Fatemi, Rohan Anil, Ethan Dyer, Siamak Shakeri, Roopali Vij, Harsh Mehta, Vinay Ramasesh, Quoc Le, Ed Chi, Yifeng Lu, Orhan Firat, et al. Michelangelo: Long Context Evaluations Beyond Haystacks via Latent Structure Queries. *arXiv preprint arXiv:2409.12640*, 2024. URL <https://arxiv.org/abs/2409.12640>.
- [206] Bing Wang, Xinnian Liang, Jian Yang, Hui Huang, Shuangzhi Wu, Peihao Wu, Lu Lu, Zejun Ma, and Zhoujun Li. SCM: Enhancing Large Language Model with Self-Controlled Memory Framework. *arXiv preprint arXiv:2304.13343*, 2023. URL <https://arxiv.org/abs/2304.13343>.
- [207] Bo Wang, Weiye He, Shenglai Zeng, Zhen Xiang, Yue Xing, Jiliang Tang, and Pengfei He. Unveiling Privacy Risks in LLM Agent Memory. *arXiv preprint arXiv:2502.13172*, 2025. URL <https://arxiv.org/abs/2502.13172>.
- [208] Chenyu Wang, Yihan Wang, and Kai Li. Evaluating Zero-Shot Long-Context LLM Compression. *arXiv preprint arXiv:2406.06773*, 2024. URL <https://arxiv.org/abs/2406.06773>.
- [209] Chonghua Wang, Haodong Duan, Songyang Zhang, Dahua Lin, and Kai Chen. Ada-LEval: Evaluating long-context LLMs with length-adaptable benchmarks. *arXiv preprint arXiv:2404.06480*, 2024. URL <https://arxiv.org/abs/2404.06480>.
- [210] Hengyi Wang, Haizhou Shi, Shiwei Tan, Weiye Qin, Wenyuan Wang, Tunyu Zhang, Akshay Nambi, Tanuja Ganu, and Hao Wang. Multimodal Needle in a Haystack: Benchmarking Long-Context Capability of Multimodal Large Language Models. *arXiv preprint arXiv:2406.11230*, 2024. URL <https://arxiv.org/abs/2406.11230>.
- [211] Lanrui Wang, Mingyu Zheng, Hongyin Tang, Zheng Lin, Yanan Cao, Jingang Wang, Xunliang Cai, and Weiping Wang. NeedleInATable: Exploring Long-Context Capability of Large Language Models towards Long-Structured Tables. *arXiv preprint arXiv:2504.06560*, 2025. URL <https://arxiv.org/abs/2504.06560>.

- [212] Lei Wang, Shan Dong, Yuhui Xu, Hanze Dong, Yalu Wang, Amrita Saha, Ee-Peng Lim, Caiming Xiong, and Doyen Sahoo. MathHay: An Automated Benchmark for Long-Context Mathematical Reasoning in LLMs. *arXiv preprint arXiv:2410.04698*, 2024. URL <https://arxiv.org/abs/2410.04698>.
- [213] Minzheng Wang, Longze Chen, Cheng Fu, Shengyi Liao, Xinghua Zhang, Bingli Wu, Haiyang Yu, Nan Xu, Lei Zhang, Run Luo, Yunshui Li, Min Yang, Fei Huang, and Yongbin Li. Leave No Document Behind: Benchmarking Long-Context LLMs with Extended Multi-Doc QA. *arXiv preprint arXiv:2406.17419*, 2024. URL <https://arxiv.org/abs/2406.17419>.
- [214] Peng Wang, Zexi Li, Ningyu Zhang, Ziwen Xu, Yunzhi Yao, Yong Jiang, Pengjun Xie, Fei Huang, and Huajun Chen. WISE: Rethinking the Knowledge Memory for Lifelong Model Editing of Large Language Models. *arXiv preprint arXiv:2405.14768*, 2024. URL <https://arxiv.org/abs/2405.14768>.
- [215] Qingyue Wang, Yanhe Fu, Yanan Cao, Shuai Wang, Zhiliang Tian, and Liang Ding. Recursively Summarizing Enables Long-Term Dialogue Memory in Large Language Models. *arXiv preprint arXiv:2308.15022*, 2023. URL <https://arxiv.org/abs/2308.15022>.
- [216] Siyuan Wang, Gaokai Zhang, Li Lyna Zhang, Ning Shang, Fan Yang, Dongyao Chen, and Mao Yang. LoongRL: Reinforcement Learning for Advanced Reasoning over Long Contexts. *arXiv preprint arXiv:2510.19363*, 2025. URL <https://arxiv.org/abs/2510.19363>.
- [217] Weilan Wang, Yu Mao, Dongdong Tang, Hongchao Du, Nan Guan, and Chun Jason Xue. When Compression Meets Model Compression: Memory-Efficient Double Compression for Large Language Models. *arXiv preprint arXiv:2502.15443*, 2025. URL <https://arxiv.org/abs/2502.15443>.
- [218] Xinghao Wang, Pengyu Wang, Bo Wang, Dong Zhang, Yunhua Zhou, and Xipeng Qiu. BitStack: Any-Size Compression of Large Language Models in Variable Memory Environments. *arXiv preprint arXiv:2410.23918*, 2024. URL <https://arxiv.org/abs/2410.23918>.
- [219] Yanting Wang, Wei Zou, Runpeng Geng, and Jinyuan Jia. TracLLM: A Generic Framework for Attributing Long Context LLMs. *arXiv preprint arXiv:2506.04202*, 2025. URL <https://arxiv.org/abs/2506.04202>.
- [220] Ziming Wang. TOKI: A Bitemporal Operator Algebra for Contradiction Resolution in LLM-Agent Persistent Memory. *arXiv preprint arXiv:2606.06240*, 2026. URL <https://arxiv.org/abs/2606.06240>.
- [221] Zixiong Wang, Gaoyang Liu, Yang Yang, and Chen Wang. Membership Inference Attack against Long-Context Large Language Models. *arXiv preprint arXiv:2411.11424*, 2024. URL <https://arxiv.org/abs/2411.11424>.
- [222] Jiale Wei, Shuchi Wu, Ruochen Liu, Xiang Ying, Jingbo Shang, and Fangbo Tao. Tuning LLMs by RAG Principles: Towards LLM-native Memory. *arXiv preprint arXiv:2503.16071*, 2025. URL <https://arxiv.org/abs/2503.16071>.
- [223] Lingxiao Wei, He Yan, Xiangju Lu, Junmin Zhu, Jun Wang, and Wei Zhang. CNNSum: Exploring Long-Context Summarization with Large Language Models in Chinese Novels. *arXiv preprint arXiv:2412.02819*, 2024. URL <https://arxiv.org/abs/2412.02819>.
- [224] Qianshan Wei, Tengchao Yang, Yaochen Wang, Xinfeng Li, Lijun Li, Zhenfei Yin, Yi Zhan, Thorsten Holz, Zhiqiang Lin, and XiaoFeng Wang. A-MemGuard: A Proactive Defense Framework for LLM-Based Agent Memory. *arXiv preprint arXiv:2510.02373*, 2025. URL <https://arxiv.org/abs/2510.02373>.
- [225] Rubin Wei, Jiaqi Cao, Jiarui Wang, Jushi Kai, Qipeng Guo, Bowen Zhou, and Zhouhan Lin. MLP Memory: A Retriever-Pretrained Memory for Large Language Models. *arXiv preprint arXiv:2508.01832*, 2025. URL <https://arxiv.org/abs/2508.01832>.
- [226] Tianxin Wei, Noveen Sachdeva, Benjamin Coleman, Zhankui He, Yuanchen Bei, Xuying Ning, Mengting Ai, Yunzhe Li, Jingrui He, Ed H. Chi, Chi Wang, Shuo Chen, Fernando Pereira, Wang-Cheng Kang, and Derek Zhiyuan Cheng. Evo-Memory: Benchmarking LLM Agent Test-time Learning with Self-Evolving Memory. *arXiv preprint arXiv:2511.20857*, 2025. URL <https://arxiv.org/abs/2511.20857>.
- [227] Longyun Wu, Dawei Zhu, Guangxiang Zhao, Zhuocheng Yu, Junfeng Ran, Xiangyu Wong, Lin Sun, and Sujian Li. LongAttn: Selecting Long-context Training Data via Token-level Attention. *arXiv preprint arXiv:2502.16860*, 2025. URL <https://arxiv.org/abs/2502.16860>.
- [228] Wenhao Wu, Yizhong Wang, Yao Fu, Xiang Yue, Dawei Zhu, and Sujian Li. Long Context Alignment with Short Instructions and Synthesized Positions. *arXiv preprint arXiv:2405.03939*, 2024. URL <https://arxiv.org/abs/2405.03939>.
- [229] Xiaodong Wu, Minhao Wang, Yichen Liu, Xiaoming Shi, He Yan, Xiangju Lu, Junmin Zhu, and Wei Zhang. LIFBench: Evaluating the Instruction Following Performance and Stability of

- Large Language Models in Long-Context Scenarios. *arXiv preprint arXiv:2411.07037*, 2024. URL <https://arxiv.org/abs/2411.07037>.
- [230] Zijun Wu, Bingyuan Liu, Ran Yan, Lei Chen, and Thomas Deitel. Reducing Distraction in Long-Context Language Models by Focused Learning. *arXiv preprint arXiv:2411.05928*, 2024. URL <https://arxiv.org/abs/2411.05928>.
- [231] Zijun Wu, Yongchang Hao, and Lili Mou. TokMem: One-Token Procedural Memory for Large Language Models. *arXiv preprint arXiv:2510.00444*, 2025. URL <https://arxiv.org/abs/2510.00444>.
- [232] Siyu Xia, Zekun Xu, Jiajun Chai, Wentian Fan, Yan Song, Xiaohan Wang, Guojun Yin, Wei Lin, Haifeng Zhang, and Jun Wang. From Experience to Strategy: Empowering LLM Agents with Trainable Graph Memory. *arXiv preprint arXiv:2511.07800*, 2025. URL <https://arxiv.org/abs/2511.07800>.
- [233] Yanling Xiao, Huaibing Xie, Guoliang Zhao, Shihan Dou, Shaolei Wang, Yiting Liu, Nantao Zheng, Cheng Zhang, Pluto Zhou, Zhisong Zhang, and Lemao Liu. A Decomposition Perspective to Long-context Reasoning for LLMs. *arXiv preprint arXiv:2604.07981*, 2026. URL <https://arxiv.org/abs/2604.07981>.
- [234] Yao Xiao, Lei Wang, Yue Deng, Guanzheng Chen, Ziqi Jin, Jung jae Kim, Xiaoli Li, Roy Ka wei Lee, and Lidong Bing. Document Reconstruction Unlocks Scalable Long-Context RLVR. *arXiv preprint arXiv:2602.08237*, 2026. URL <https://arxiv.org/abs/2602.08237>.
- [235] Huaibing Xie, Guoliang Zhao, Yang Liu, Shihan Dou, Siming Huang, Yanling Xiao, Shaolei Wang, Yiting Liu, Cheng Zhang, Shaofan Liu, and Pluto Zhou. Probing How Scalable Table Data Enhances General Long-Context Reasoning. *arXiv preprint arXiv:2603.21719*, 2026. URL <https://arxiv.org/abs/2603.21719>.
- [236] Rui Xie, Asad Ul Haq, Linsen Ma, Yunhua Fang, Zirak Burzin Engineer, Liu Liu, and Tong Zhang. Reimagining Memory Access for LLM Inference: Compression-Aware Memory Controller Design. *arXiv preprint arXiv:2503.18869*, 2025. URL <https://arxiv.org/abs/2503.18869>.
- [237] Weiwei Xie, Shaoxiong Guo, Fan Zhang, Tian Xia, Xue Yang, Lizhuang Ma, Junchi Yan, and Qibing Ren. MemEvoBench: Benchmarking Safety Risks from Memory Misevolution in LLM Agents. *arXiv preprint arXiv:2604.15774*, 2026. URL <https://arxiv.org/abs/2604.15774>.
- [238] Yue Xing, Tao Yang, Yijiashun Qi, Minggu Wei, Yu Cheng, and Honghui Xin. Structured Memory Mechanisms for Stable Context Representation in Large Language Models. *arXiv preprint arXiv:2505.22921*, 2025. URL <https://arxiv.org/abs/2505.22921>.
- [239] Hua-Dong Xiong, Li Ji-An, Jiaqi Huang, Robert C. Wilson, Kwonjoon Lee, and Xue-Xin Wei. Human-like Working Memory Interference in Large Language Models. *arXiv preprint arXiv:2604.09670*, 2026. URL <https://arxiv.org/abs/2604.09670>.
- [240] Qiushi Xiong, Zhipeng Xu, Zhenghao Liu, Mengjia Wang, Zulong Chen, Yue Sun, Yu Gu, Xiaohua Li, and Ge Yu. Enhancing the Patent Matching Capability of Large Language Models via the Memory Graph. *arXiv preprint arXiv:2504.14845*, 2025. URL <https://arxiv.org/abs/2504.14845>.
- [241] Zidi Xiong, Yuping Lin, Wenya Xie, Pengfei He, Zirui Liu, Jiliang Tang, Himabindu Lakkaraju, and Zhen Xiang. How Memory Management Impacts LLM Agents: An Empirical Study of Experience-Following Behavior. *arXiv preprint arXiv:2505.16067*, 2025. URL <https://arxiv.org/abs/2505.16067>.
- [242] Jingyan Xu, Hong Shi, Yi Shan, Penghui Liu, Yunhao Bai, Ningyuan Li, and Xueyang Liu. TrustMargin: Training-Free Arbitration between Parametric Memory and Retrieved Evidence in Large Language Models. *arXiv preprint arXiv:2606.08397*, 2026. URL <https://arxiv.org/abs/2606.08397>.
- [243] Peng Xu, Wei Ping, Xianchao Wu, Lawrence McAfee, Chen Zhu, Zihan Liu, Sandeep Subramanian, Evelina Bakhturina, Mohammad Shoeybi, and Bryan Catanzaro. Retrieval meets Long Context Large Language Models. *arXiv preprint arXiv:2310.03025*, 2023. URL <https://arxiv.org/abs/2310.03025>.
- [244] Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. A-MEM: Agentic Memory for LLM Agents. *arXiv preprint arXiv:2502.121101*, 2025. URL <https://arxiv.org/abs/2502.121101>.
- [245] Xingzi Xu, Amir Tavaneai, Kavosh Asadi, and Karim Bouyarmane. Adjoint sharding for very long context training of state space models. *arXiv preprint arXiv:2501.00692*, 2025. URL <https://arxiv.org/abs/2501.00692>.
- [246] Xun Xu. G-MemLLM: Gated Latent Memory Augmentation for Long-Context Reasoning in Large Language Models. *arXiv preprint arXiv:2602.00015*, 2026. URL <https://arxiv.org/abs/2602.00015>.
- [247] Yi Xu, Ziming Mao, Xiangxi Mo, Shu Liu, and Ion Stoica. Pie: Pooling CPU Memory for LLM Inference. *arXiv preprint arXiv:2411.09317*, 2024. URL <https://arxiv.org/abs/2411.09317>.
- [248] Zhe Xu, Jiasheng Ye, Xiaoran Liu, Xiangyang Liu, Tianxiang Sun, Zhigeng Liu, Qipeng Guo, Linlin

- Li, Qun Liu, Xuanjing Huang, and Xipeng Qiu. DetectiveQA: Evaluating Long-Context Reasoning on Detective Novels. *arXiv preprint arXiv:2409.02465*, 2024. URL <https://arxiv.org/abs/2409.02465>.
- [249] Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Jinhe Bi, Kristian Kersting, Jeff Z. Pan, Hinrich Schutze, Volker Tresp, and Yunpu Ma. Memory-R1: Enhancing Large Language Model Agents to Manage and Utilize Memories via Reinforcement Learning. *arXiv preprint arXiv:2508.19828*, 2025. URL <https://arxiv.org/abs/2508.19828>.
- [250] Sikuan Yan, Ahmed Bahloul, Ercong Nie, Susanna Schwarzmann, Riccardo Trivisonno, Volker Tresp, and Yunpu Ma. Memory-R2: Fair Credit Assignment for Long-Horizon Memory-Augmented LLM Agents. *arXiv preprint arXiv:2605.21768*, 2026. URL <https://arxiv.org/abs/2605.21768>.
- [251] Yuchen Yan, Yongliang Shen, Yang Liu, Jin Jiang, Mengdi Zhang, Jian Shao, and Yueting Zhuang. InftyThink: Breaking the Length Limits of Long-Context Reasoning in Large Language Models. *arXiv preprint arXiv:2503.06692*, 2025. URL <https://arxiv.org/abs/2503.06692>.
- [252] Hao Yang, Shiqi Shen, Haoxuan Li, Zhipeng Wang, Zhi Gong, and Xu Chen. Rosetta Memory: Adaptive Memory for Cross-LLM Agents. *arXiv preprint arXiv:2606.07711*, 2026. URL <https://arxiv.org/abs/2606.07711>.
- [253] Wei Yang, Jinwei Xiao, Hongming Zhang, Qingyang Zhang, Yanna Wang, and Bo Xu. Coarse-to-Fine Grounded Memory for LLM Agent Planning. *arXiv preprint arXiv:2508.15305*, 2025. URL <https://arxiv.org/abs/2508.15305>.
- [254] Jinghan Yao, Sam Ade Jacobs, Masahiro Tanaka, Olatunji Ruwase, Hari Subramoni, and Dhavalbaleswar K. Panda. Training Ultra Long Context Language Model with Fully Pipelined Distributed Transformer. *arXiv preprint arXiv:2408.16978*, 2024. URL <https://arxiv.org/abs/2408.16978>.
- [255] Inseok Yeo, Duksan Ryu, and Jongmoon Baik. Improving LLM-Based Fault Localization with External Memory and Project Context. *arXiv preprint arXiv:2506.03585*, 2025. URL <https://arxiv.org/abs/2506.03585>.
- [256] Xingkun Yin and Hongyang Du. GLOVE: Global Verifier for LLM Memory-Environment Realignment. *arXiv preprint arXiv:2601.19249*, 2026. URL <https://arxiv.org/abs/2601.19249>.
- [257] Song Yu, Xiaofei Xu, Ke Deng, Li Li, and Lin Tian. Tree of Agents: Improving Long-Context Capabilities of Large Language Models through Multi-Perspective Reasoning. *arXiv preprint arXiv:2509.06436*, 2025. URL <https://arxiv.org/abs/2509.06436>.
- [258] Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyang Deng, Yupeng Cao, Zhi Chen, Jordan W. Suchow, Rong Liu, Zhenyu Cui, Zhaozhao Xu, Denghui Zhang, Koduvayur Subbalakshmi, Guojun Xiong, Yueru He, Jimin Huang, Dong Li, and Qianqian Xie. FinCon: A Synthesized LLM Multi-Agent System with Conceptual Verbal Reinforcement for Enhanced Financial Decision Making. *arXiv preprint arXiv:2407.06567*, 2024. URL <https://arxiv.org/abs/2407.06567>.
- [259] Yi Yu, Liuyi Yao, Yuexiang Xie, Qingquan Tan, Jiaqi Feng, Yaliang Li, and Libing Wu. Agentic Memory: Learning Unified Long-Term and Short-Term Memory Management for Large Language Model Agents. *arXiv preprint arXiv:2601.01885*, 2026. URL <https://arxiv.org/abs/2601.01885>.
- [260] Yijiong Yu. An Effective Framework to Help Large Language Models Handle Numeric-involved Long-context Tasks. *arXiv preprint arXiv:2411.10145*, 2024. URL <https://arxiv.org/abs/2411.10145>.
- [261] Yijiong Yu, Yongfeng Huang, Zhixiao Qi, and Zhe Zhou. Training With "Paraphrasing the Original Text" Teaches LLM to Better Retrieve in Long-context Tasks. *arXiv preprint arXiv:2312.111930*, 2023. URL <https://arxiv.org/abs/2312.111930>.
- [262] Boqin Yuan, Yue Su, and Kun Yao. Diagnosing Retrieval vs. Utilization Bottlenecks in LLM Agent Memory. *arXiv preprint arXiv:2603.02473*, 2026. URL <https://arxiv.org/abs/2603.02473>.
- [263] Ruifeng Yuan, Shichao Sun, Yongqi Li, Zili Wang, Ziqiang Cao, and Wenjie Li. Personalized Large Language Model Assistant with Evolving Conditional Memory. *arXiv preprint arXiv:2312.17257*, 2023. URL <https://arxiv.org/abs/2312.17257>.
- [264] Xihang Yue, Linchao Zhu, and Yi Yang. FragRel: Exploiting Fragment-level Relations in the External Memory of Large Language Models. *arXiv preprint arXiv:2406.03092*, 2024. URL <https://arxiv.org/abs/2406.03092>.
- [265] Yuxuan Yue, Zhihang Yuan, Haojie Duanmu, Sifan Zhou, Jianlong Wu, and Liqiang Nie. WKVQuant: Quantizing Weight and Key/Value Cache for Large Language Models Gains More. *arXiv preprint arXiv:2402.12065*, 2024. URL <https://arxiv.org/abs/2402.12065>.
- [266] Sizhe Yuen, Francisco Gomez Medina, Ting Su, Yali Du, and Adam J. Sobey. Intrinsic Memory Agents: Heterogeneous Multi-Agent LLM Systems through Structured Contextual Memory. *arXiv preprint arXiv:2508.08997*, 2025. URL <https://arxiv.org/abs/2508.08997>.
- [267] Ruihong Zeng, Jinyuan Fang, Siwei Liu, and Zaiqiao Meng. On the Structural Memory of LLM Agents. *arXiv preprint arXiv:2412.15266*, 2024. URL <https://arxiv.org/abs/2412.15266>.

- [268] Zhen Zeng, Leijiang Gu, Zhangling Duan, Feng Li, Cees G. M. Snoek, Meng Wang, and Zenglin Shi. Towards Benign Memory Forgetting for Selective Multimodal Large Language Model Unlearning. *arXiv preprint arXiv:2511.20196*, 2025. URL <https://arxiv.org/abs/2511.20196>.
- [269] Stefano Zeppieri. MMAG: Mixed Memory-Augmented Generation for Large Language Models Applications. *arXiv preprint arXiv:2512.01710*, 2025. URL <https://arxiv.org/abs/2512.01710>.
- [270] Chen Zhang, Kuntai Du, Shu Liu, Woosuk Kwon, Xiangxi Mo, Yufeng Wang, Xiaoxuan Liu, Kaichao You, Zhuohan Li, Mingsheng Long, Jidong Zhai, Joseph Gonzalez, and Ion Stoica. Jenga: Effective Memory Management for Serving LLM with Heterogeneity. *arXiv preprint arXiv:2503.18292*, 2025. URL <https://arxiv.org/abs/2503.18292>.
- [271] Dianxing Zhang, Wendong Li, Kani Song, Jiaye Lu, Gang Li, Liuchun Yang, and Sheng Li. Memory in Large Language Models: Mechanisms, Evaluation and Evolution. *arXiv preprint arXiv:2509.18868*, 2025. URL <https://arxiv.org/abs/2509.18868>.
- [272] Guilin Zhang, Wei Jiang, Xiejiashan Wang, Aisha Behr, Kai Zhao, Jeffrey Friedman, Xu Chu, and Amine Anoun. Adaptive Memory Admission Control for LLM Agents. *arXiv preprint arXiv:2603.04549*, 2026. URL <https://arxiv.org/abs/2603.04549>.
- [273] Hanzhi Zhang, Heng Fan, Kewei Sha, Yan Huang, and Yunhe Feng. DAM: Dynamic Attention Mask for Long-Context Large Language Model Inference Acceleration. *arXiv preprint arXiv:2506.11104*, 2025. URL <https://arxiv.org/abs/2506.11104>.
- [274] Haozhen Zhang, Tao Feng, Pengrui Han, and Jiaxuan You. AcademicEval: Live Long-Context LLM Benchmark. *arXiv preprint arXiv:2510.17725*, 2025. URL <https://arxiv.org/abs/2510.17725>.
- [275] Jinrui Zhang, Chaodong Xiao, Aoqi Wu, Xindong Zhang, and Lei Zhang. Pretraining A Large Language Model using Distributed GPUs: A Memory-Efficient Decentralized Paradigm. *arXiv preprint arXiv:2602.11543*, 2026. URL <https://arxiv.org/abs/2602.11543>.
- [276] Lei Zhang, Yunshui Li, Ziqiang Liu, Jiayi Yang, Junhao Liu, Longze Chen, Run Luo, and Min Yang. Marathon: A Race Through the Realm of Long Context with Large Language Models. *arXiv preprint arXiv:2312.09542*, 2023. URL <https://arxiv.org/abs/2312.09542>.
- [277] Shaohua Zhang, Yuan Lin, and Hang Li. Memory Retrieval and Consolidation in Large Language Models through Function Tokens. *arXiv preprint arXiv:2510.08203*, 2025. URL <https://arxiv.org/abs/2510.08203>.
- [278] Sheng Zhang, Junyi Li, Yingyi Zhang, Pengyue Jia, Yichao Wang, Xiaowei Qian, Wenlin Zhang, Maolin Wang, Yong Liu, and Xiangyu Zhao. MemSearch-o1: Empowering Large Language Models with Reasoning-Aligned Memory Growth in Agentic Search. *arXiv preprint arXiv:2604.17265*, 2026. URL <https://arxiv.org/abs/2604.17265>.
- [279] Shuning Zhang, Rongjun Ma, Ying Ma, Shixuan Li, Yiqun Xu, Xin Yi, and Hewu Li. Understanding Users’ Privacy Perceptions Towards LLM’s RAG-based Memory. *arXiv preprint arXiv:2508.07664*, 2025. URL <https://arxiv.org/abs/2508.07664>.
- [280] Tianhua Zhang, Xinjiang Wang, Qianxi Zhang, Qi Chen, Kun Li, Yaoqi Chen, Dingdong Wang, Helen Meng, and Yan Lu. TriggerBench: Investigating Prospective Memory for Large Language Models. *arXiv preprint arXiv:2606.23459*, 2026. URL <https://arxiv.org/abs/2606.23459>.
- [281] Xin Zhang, Kailai Yang, Hao Li, Chenyue Li, Qiyu Wei, and Sophia Ananiadou. Implicit Graph, Explicit Retrieval: Towards Efficient and Interpretable Long-horizon Memory for Large Language Models. *arXiv preprint arXiv:2601.03417*, 2026. URL <https://arxiv.org/abs/2601.03417>.
- [282] Xinliang Frederick Zhang, Nick Beauchamp, and Lu Wang. PRIME: Large Language Model Personalization with Cognitive Dual-Memory and Personalized Thought Process. *arXiv preprint arXiv:2507.04607*, 2025. URL <https://arxiv.org/abs/2507.04607>.
- [283] Xinrong Zhang, Yingfa Chen, Shengding Hu, Zihang Xu, Junhao Chen, Moo Khai Hao, Xu Han, Zhen Leng Thai, Shuo Wang, Zhiyuan Liu, and Maosong Sun. \$infty\$Bench: Extending Long Context Evaluation Beyond 100K Tokens. *arXiv preprint arXiv:2402.13718*, 2024. URL <https://arxiv.org/abs/2402.13718>.
- [284] Xinsen Zhang, Zhenkai Ding, Tianjun Pan, Run Yang, Chun Kang, Xue Xiong, and Jingnan Gu. OPSDL: On-Policy Self-Distillation for Long-Context Language Models. *arXiv preprint arXiv:2604.17535*, 2026. URL <https://arxiv.org/abs/2604.17535>.
- [285] Yiran Zhang, Jincheng Hu, Mark Dras, and Usman Naseem. CogMem: A Cognitive Memory Architecture for Sustained Multi-Turn Reasoning in Large Language Models. *arXiv preprint arXiv:2512.14118*, 2025. URL <https://arxiv.org/abs/2512.14118>.
- [286] Yusen Zhang, Ruoxi Sun, Yanfei Chen, Tomas Pfister, Rui Zhang, and Sercan O. Arik. Chain of Agents: Large Language Models Collaborating on Long-Context Tasks. *arXiv preprint arXiv:2406.02818*, 2024. URL <https://arxiv.org/abs/2406.02818>.

- [287] Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. A Survey on the Memory Mechanism of Large Language Model based Agents. *arXiv preprint arXiv:2404.13501*, 2024. URL <https://arxiv.org/abs/2404.13501>.
- [288] Zeyu Zhang, Quanyu Dai, Rui Li, Xiaohe Bo, Xu Chen, and Zhenhua Dong. Learn to Memorize: Optimizing LLM-based Agents with Adaptive Memory Framework. *arXiv preprint arXiv:2508.16629*, 2025. URL <https://arxiv.org/abs/2508.16629>.
- [289] Zhihan Zhang, Yixin Cao, Chenchen Ye, Yunshan Ma, Lizi Liao, and Tat-Seng Chua. Analyzing Temporal Complex Events with Large Language Models? A Benchmark towards Temporal, Long Context Understanding. *arXiv preprint arXiv:2406.02472*, 2024. URL <https://arxiv.org/abs/2406.02472>.
- [290] Haiquan Zhao, Xuwu Wang, Shisong Chen, Zhixu Li, Xin Zheng, and Yanghua Xiao. OVEL: Large Language Model as Memory Manager for Online Video Entity Linking. *arXiv preprint arXiv:2403.01411*, 2024. URL <https://arxiv.org/abs/2403.01411>.
- [291] Jiachen Zhao. In-Context Exemplars as Clues to Retrieving from Large Associative Memory. *arXiv preprint arXiv:2311.03498*, 2023. URL <https://arxiv.org/abs/2311.03498>.
- [292] Qingfei Zhao, Ruobing Wang, Yukuo Cen, Daren Zha, Shicheng Tan, Yuxiao Dong, and Jie Tang. LongRAG: A Dual-Perspective Retrieval-Augmented Generation Paradigm for Long-Context Question Answering. *arXiv preprint arXiv:2410.18050*, 2024. URL <https://arxiv.org/abs/2410.18050>.
- [293] Yunao Zheng, Xiaojie Wang, Lei Ren, and Wei Chen. ROSA-Tuning: Enhancing Long-Context Modeling via Suffix Matching. *arXiv preprint arXiv:2602.02499*, 2026. URL <https://arxiv.org/abs/2602.02499>.
- [294] Chenyu Zhou, Huacan Chai, Wenteng Chen, Zihan Guo, Rong Shan, Yuanyi Song, Tianyi Xu, Yingxuan Yang, Aofan Yu, Weiming Zhang, Congming Zheng, Jiachen Zhu, Zeyu Zheng, Zhuosheng Zhang, Xingyu Lou, Changwang Zhang, Zhihui Fu, Jun Wang, Weiwen Liu, Jianghao Lin, et al. Externalization in LLM Agents: A Unified Review of Memory, Skills, Protocols and Harness Engineering. *arXiv preprint arXiv:2604.08224*, 2026. URL <https://arxiv.org/abs/2604.08224>.
- [295] Wei Zhou, Xuanhe Zhou, Shaokun Han, Hongming Xu, Guoliang Li, Zhiyu Li, Feiyu Xiong, and Fan Wu. Are We Ready For An Agent-Native Memory System? *arXiv preprint arXiv:2606.24775*, 2026. URL <https://arxiv.org/abs/2606.24775>.
- [296] Dawei Zhu, Xiyu Wei, Guangxiang Zhao, Wenhao Wu, Haosheng Zou, Junfeng Ran, Xun Wang, Lin Sun, Xiangzheng Zhang, and Sujian Li. Chain-of-Thought Matters: Improving Long-Context Language Models with Reasoning Path Supervision. *arXiv preprint arXiv:2502.20790*, 2025. URL <https://arxiv.org/abs/2502.20790>.
- [297] Wenhao Zhu, Pinzhen Chen, Hanxu Hu, Shujian Huang, Fei Yuan, Jiajun Chen, and Alexandra Birch. Generalizing From Short to Long: Effective Data Synthesis for Long-Context Instruction Tuning. *arXiv preprint arXiv:2502.15592*, 2025. URL <https://arxiv.org/abs/2502.15592>.
- [298] Yuanjie Zhu, Liangwei Yang, Ke Xu, Weizhi Zhang, Ziheng Song, Jindong Wang, and Philip S. Yu. LLM-MemCluster: Empowering Large Language Models with Dynamic Memory for Text Clustering. *arXiv preprint arXiv:2511.15424*, 2025. URL <https://arxiv.org/abs/2511.15424>.
- [299] Yufan Zhuang, Xiaodong Yu, Jialian Wu, Ximeng Sun, Ze Wang, Jiang Liu, Yusheng Su, Jingbo Shang, Zicheng Liu, and Emad Barsoum. Self-Taught Agentic Long Context Understanding. *arXiv preprint arXiv:2502.15920*, 2025. URL <https://arxiv.org/abs/2502.15920>.
- [300] Wazeer Zulfikar, Samantha Chan, and Pattie Maes. Memoro: Using Large Language Models to Realize a Concise Interface for Real-Time Memory Augmentation. *arXiv preprint arXiv:2403.02135*, 2024. URL <https://arxiv.org/abs/2403.02135>.